

Software Development Company Website

Product Requirements Document

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This document defines a substantial corporate website for a custom software development and digital product engineering company. It provides the scope, content, interaction rules, data contracts, publishing controls and release tests needed to design, build and operate the website. The recommended implementation is a modular Next.js website with a managed editorial CMS and a separate, small operational database for enquiries. The planned information architecture contains 79 named routes or reusable detail templates. Publication depends on genuine capabilities and approved content, not a page-count target.

The business owner supplies and approves the company facts identified in brackets. Proposed operating practices and sample copy become publishable only after the responsible owner confirms that the company actually follows them. A professional website supports transparent evaluation; it does not guarantee bank, payment-provider or vendor approval.

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1 Document Overview

Audience: business owner, product manager, designer, frontend and backend engineers, content editor, SEO specialist, QA engineer, security reviewer and operations owner.

Status: implementation specification with business-fact dependencies. The document recommends product decisions; it does not establish that any particular legal entity, service capability, certification, employee or customer exists.

Document conventions: MUST is a launch acceptance requirement when its feature is in scope; SHOULD is the recommended implementation; MAY is optional. P0 means required for launch, P1 means the next planned release and P2 means a separately funded future feature. A P0 conditional feature is required only when the corresponding business activity is offered. REQ identifiers and AC identifiers support ticket and test traceability.

Approvals: product owner approves scope; engineering lead approves architecture; design lead approves accessible layouts; business owner verifies claims; counsel approves applicable policies; security lead approves controls; QA owns release evidence. One person may fill several roles, but each responsibility must have a named owner.

Change control: maintain a versioned decision log with decision, date, owner, rationale, affected requirements and acceptance tests. Changes to public company information, tracking, uploads, authentication or commercial terms require the relevant owner to review the change before publication.

2 Executive Summary

Build a website that lets a buyer answer three questions: can this company handle my problem, how will the work be delivered, and what happens if I start a conversation? Organize the offer into product strategy and design; application engineering; AI and integration; cloud and modernization; and quality, support and teams.

Prioritize substantive service pages, transparent engagement terms, an understandable development process and reliable enquiry handling. Use real project evidence where available. When evidence is absent, explain deliverables and decisions using clearly labelled illustrative diagrams rather than invented client outcomes.

P0 includes the full reusable design system, core corporate and policy pages, approved service pages, content management, contact flow, security, accessibility and monitoring. The 79-route plan is the supported target architecture, not a requirement to publish unsupported capabilities. P1 adds the guided enquiry, richer editorial search and additional verified sector content. P2 includes the client portal and multilingual content.

3 Business Context

Business definition: [BRAND NAME] provides [VERIFIED SOFTWARE SERVICES] through [ACTUAL DELIVERY MODEL]. The contracting entity is [LEGAL COMPANY NAME], registered in [JURISDICTION]. Its public operating locations, registered address and delivery coverage must be distinguished explicitly.

Software engagements are typically evaluated across business fit, architecture, commercial risk and support. The website must therefore serve early research as well as detailed procurement review. A

visually complete site without a working intake process or supported claims fails the product objective.

Assumptions: English first; B2B enquiries; no public checkout; no automatic project prices; no customer login at launch; a small marketing team maintains content; clients receive proposals and contracts outside the marketing website. These are recommendations and may be changed through the decision log.

Exclusions: payment processing, customer banking, resale of third-party subscriptions, self-service SaaS billing, public AI chat, authenticated client collaboration and promises about regulated-sector compliance. Any actual additional business activities must be disclosed accurately rather than hidden to change how the company is classified.

4 Product Vision

Create a dependable reference point for understanding the company and starting an informed software project. Buyers should leave with a clearer view of scope, delivery decisions, responsibilities, ownership and ongoing costs.

The interface should feel composed and technically literate: readable typography, original diagrams, real screenshots, generous space and useful answers. Visual sophistication must come from design quality and content depth rather than exaggerated scale or manufactured social proof.

5 Website Objectives

Objective	Measurement	Launch or operating criterion
Explain the offer	Moderated task test	At least 4 of 5 representative participants identify services and next step without coaching
Explain engineering	Process and technology task test	CTO participant finds testing, deployment and ownership information within three navigation choices
Generate leads	Valid enquiries and qualified enquiries	Establish first 30-day baseline; set growth targets only after measuring traffic quality
Enable evaluation	Engagement and service content coverage	Every published commercial page states deliverables, boundaries and CTA
Support due diligence	Identity and policy audit	No conflicting legal name, contact, location or billing information
Enable organic discovery	Indexable page quality and search reporting	All intended pages crawlable, self-canonical and included in sitemap; no ranking promise

Metric definitions: enquiry conversion = unique accepted project enquiries divided by measurable eligible sessions in the same period; qualified rate = enquiries meeting approved service, budget and timing criteria divided by reviewed non-spam enquiries. Report consent-related measurement gaps. Count enquiries from operational records even when analytics is disabled; do not use those records to bypass tracking consent.

6 Target Audience

Primary audiences are founders, product owners and technology leaders with a concrete software initiative. Secondary audiences include operations, marketing technology, procurement, job seekers and delivery partners. Initial service and industry priorities must follow actual sales demand and staff capability.

Primary journeys: problem → solution → service → process → enquiry; technical evaluation → technology → security → case evidence → technical discussion; commercial evaluation → engagement models → ownership and support → enquiry; due diligence → company identity → policies → contact; candidate → careers → genuine vacancy → application.

7 Buyer Personas

Startup founder

Problem: uncertain scope, constrained runway and limited technical capacity. Motivation: validate a useful first release before committing to a large build. Objections: scope inflation, lock-in and unclear ownership. Decision criteria: discovery outputs, staged investment and maintainability. Expected information: what an MVP includes, exclusions, founder responsibilities and handover. CTA: “Plan Your MVP.” Confidence content: sample discovery agenda, acceptance checklist and a real early-stage project when available.

SaaS founder

Problem: subscriptions, onboarding, tenancy and product iteration compete for attention. Motivation: launch or improve a recurring-use product. Objections: unreliable billing, weak tenant isolation and growing infrastructure costs. Criteria: entitlement design, migration approach and release process. Expected information: architecture options, integrations, maintenance and operational costs. CTA: “Discuss Your SaaS Product.” Confidence content: tenancy decision guide and verified SaaS architecture narrative.

Small business owner

Problem: disconnected tools and recurring manual work. Motivation: make one important workflow easier. Objections: custom software cost and dependence on a supplier. Criteria: build-versus-buy advice, understandable scope and ongoing costs. Expected information: practical examples, training, support and who owns accounts. CTA: “Discuss Your Workflow.” Confidence content: workflow map and plain-English engagement explanation.

SME decision maker

Problem: departments use inconsistent data and undocumented processes. Motivation: improve coordination while protecting day-to-day operations. Objections: migration disruption and employee adoption. Criteria: phased rollout, permissions and training. Expected information: integration planning, milestones and stakeholder participation. CTA: “Plan a Business System.” Confidence content: migration checklist and acceptance workshop outline.

Enterprise technology team

Problem: dependencies, vendor governance and integration complexity. Motivation: add delivery capacity for a defined initiative. Objections: procurement friction, data exposure and unsupported architecture. Criteria: documented interfaces, testing evidence and auditability. Expected information: delivery governance, security practices and contractual responsibilities. CTA: "Discuss Enterprise Delivery." Confidence content: example documentation contents and security contact.

CTO

Problem: architecture trade-offs and technical debt under delivery pressure. Motivation: obtain sound engineering decisions and an accountable partner. Objections: opaque code quality and technology hype. Criteria: architecture rationale, review process and maintainability. Expected information: stack selection, CI/CD, observability and ownership. CTA: "Discuss the Architecture." Confidence content: annotated reference architecture clearly identified as illustrative.

CIO

Problem: portfolio cost, supplier risk and fragmented systems. Motivation: create a manageable modernization programme. Objections: business interruption and incomplete governance. Criteria: phased investment, operational ownership and risk management. Expected information: delivery model, business continuity and escalation. CTA: "Plan a Modernization Discussion." Confidence content: dependency map and change governance example.

Head of engineering

Problem: hiring constraints, release bottlenecks and inconsistent contributions. Motivation: strengthen delivery without losing technical control. Objections: onboarding overhead and quality variance. Criteria: verified skills, review standards and availability. Expected information: team structure, overlap hours and replacement process. CTA: "Discuss Team Capacity." Confidence content: onboarding plan and actual role profiles.

Product manager

Problem: stakeholder disagreement and unclear acceptance criteria. Motivation: turn user needs into a sequenced release. Objections: missed requirements and late design changes. Criteria: discovery, backlog clarity and feedback cadence. Expected information: user stories, demos, UAT and change handling. CTA: "Plan Product Discovery." Confidence content: sample story and release acceptance template.

Digital transformation manager

Problem: legacy processes resist organization-wide change. Motivation: modernize a measurable business workflow. Objections: technology-first proposals and poor adoption. Criteria: process research, incremental deployment and training. Expected information: current-state mapping, stakeholder plan and migration controls. CTA: "Map Your Transformation." Confidence content: roadmap example with business dependencies.

Marketing technology team

Problem: fragmented CRM, forms, attribution and content systems. Motivation: connect tools and improve reliable data flow. Objections: tracking compliance and broken campaigns. Criteria: integration ownership, consent-aware events and reconciliation. Expected information: API limitations, analytics design and rollback. CTA: "Discuss Your Integrations." Confidence content: integration contract and event dictionary examples.

Operations team

Problem: repetitive entry, exception queues and spreadsheet dependence. Motivation: reduce avoidable manual handling. Objections: automation failures and loss of control. Criteria: approvals, human fallback and audit logs. Expected information: triggers, failure handling and access rules. CTA: "Review an Operational Workflow." Confidence content: exception-handling flow and administrator screen concept.

White-label agency

Problem: client commitments exceed engineering capacity. Motivation: obtain dependable delivery under agreed branding. Objections: confidentiality, client ownership and inconsistent communications. Criteria: non-solicitation terms if agreed, escalation and handover standards. Expected information: collaboration model, contracting entity and boundaries. CTA: "Discuss Delivery Partnership." Confidence content: sample roles and responsibilities matrix; no fictional partnership logos.

Buyer of dedicated developers

Problem: sustained capacity gap. Motivation: add specific skills without building an entire local team. Objections: hidden subcontracting, unverified CVs and turnover. Criteria: actual experience, availability and transparent management. Expected information: interview process, replacement notice and billing basis. CTA: "Discuss Developer Requirements." Confidence content: verified profiles only and onboarding checklist.

Legacy software owner

Problem: fragile releases, unsupported dependencies and inaccessible business logic. Motivation: improve the system while keeping it operational. Objections: big-bang rewrites and data loss. Criteria: assessment quality, incremental migration and reversibility. Expected information: discovery access, test strategy and cutover planning. CTA: "Review Your Existing System." Confidence content: modernization decision tree and rollback plan example.

8 Brand Positioning

Positioning: Custom Software Development & Digital Product Engineering Partner.

Core statement: "Software built around your business." Supporting copy: "Plan, design and develop software around the work your users need to do. From a first product release to an existing platform, define the scope, engineering decisions and support model before development begins."

This copy is usable without fabricated scale claims. Statements that use "we" to describe delivery practices require business approval. Prefer concrete verbs: map, design, implement, test, release and maintain. Avoid "world-class," "best," "guaranteed," arbitrary client counts and unsupported "enterprise-grade" claims.

The five service groups are Strategy and Product Design; Application Engineering; AI and Connected Workflows; Cloud and Modernization; Quality and Delivery Capacity. Use these groups in navigation and homepage cards while preserving the individual service URLs.

9 Value Proposition

Recommended copy: “Start with the problem, make the trade-offs visible, and build a product your team can operate.” Explain scope, dependencies, responsibilities and acceptance criteria together. Present maintainability as a design objective, not an outcome guarantee.

Proposed differentiators must correspond to operating artifacts: discovery produces a prioritized brief; architecture review records decisions; sprint reviews demonstrate completed work; testing produces release evidence; handover includes repositories, environment documentation and known limitations. If an artifact is not provided in the engagement, remove the associated promise.

10 Information Architecture

Services describe work the company performs. Solutions describe systems buyers need. Industries describe sector-specific workflows and constraints. Technologies explain implementation choices. These must not become near-identical pages with substituted keywords.

Use 14 principal templates: homepage, category hub, service, solution, industry, technology category, case-study listing, case-study detail, corporate/process, conversion form, resource listing, article/guide, careers/job and policy/trust. Jobs and applications share a page shell but separate validation. Reuse components without forcing every page into the same visual rhythm.

Every published page must have a useful onward path: parent hub, relevant related page and contextual contact CTA. Critical policy pages must remain reachable directly from the footer. Breadcrumbs show canonical hierarchy, not a visitor’s click history.

11 Full Sitemap

The following inventory contains 79 planned routes or detail-template slots. Dynamic collections can later create more URLs. Anchors and redirects are not additional pages. P0 service entries remain subject to verification; if fewer capabilities are real, reduce published scope rather than inventing them.

IDs	Routes or template slots	Priority
01-02	/ ; /services/	P0
03-07	/services/custom-software-development/ ; /services/web-development/ ; /services/mobile-app-development/ ; /services/saas-development/ ; /services/mvp-development/	P0 if offered
08-12	/services/ai-development/ ; /services/ai-integration/ ; /services/automation-development/ ; /services/cloud-development/ ; /services/devops/	P1; promote verified core offers

IDs	Routes or template slots	Priority
13-17	/services/api-development-integration/ ; /services/dedicated-teams/ ; /services/ui-ux-design/ ; /services/qa-testing/ ; /services/software-maintenance/	P0 if offered
18-21	/services/legacy-modernization/ ; /services/ecommerce-development/ ; /services/enterprise-software/ ; /services/technical-consulting/	P1; promote verified core offers
22	/solutions/	P0
23-28	/solutions/startup-products/ ; /solutions/digital-transformation/ ; /solutions/business-automation/ ; /solutions/customer-portals/ ; /solutions/internal-tools/ ; /solutions/crm/	P1
29-34	/solutions/erp/ ; /solutions/saas-platforms/ ; /solutions/marketplaces/ ; /solutions/ai-applications/ ; /solutions/analytics-dashboards/ ; /solutions/workflow-automation/	P1
35	/industries/	P0 if meaningful sectors verified
36-42	/industries/technology/ ; /industries/saas/ ; /industries/ecommerce/ ; /industries/professional-services/ ; /industries/fintech/ ; /industries/education/ ; /industries/healthcare-technology/	P1, sector review required
43-48	/industries/real-estate/ ; /industries/logistics/ ; /industries/travel/ ; /industries/media/ ; /industries/retail/ ; /industries/manufacturing/	P1
49-53	/technologies/ ; /technologies/frontend/ ; /technologies/backend/ ; /technologies/mobile/ ; /technologies/databases/	Hub P0; categories P1
54-58	/technologies/cloud/ ; /technologies/devops/ ; /technologies/ai-ml/ ; /technologies/automation/ ; /technologies/integrations/	P1
59-60	/case-studies/ ; /case-studies/{slug}/	P0 templates; publish only real approved work
61-65	/how-we-work/ ; /engagement-models/ ; /support/ ; /company/ ; /company/business-information/	P0
66-70	/contact/ ; /request-a-quote/ ; /resources/ ; /blog/ ; /blog/{slug}/	Quote P1; other templates P0, listings gated

IDs	Routes or template slots	Priority
71-74	/careers/ ; /careers/{slug}/ ; /trust/ ; /security/	Careers P0 truthful empty state; job conditional; trust P1; security P0
75-79	/privacy/ ; /terms/ ; /cookies/ ; /accessibility/ ; /data-processing/	P0 applicable policies; data processing P1 unless needed at launch

Guides, engineering articles and glossary entries use the resource taxonomy and article template rather than creating thin duplicate hubs. FAQ lives at /resources/#faq and on contextual pages. Team, mission and history live at /company/#team and related anchors. Contract, IP, cancellation and third-party cost explanations live at /engagement-models/ with policy links. An AUP becomes an additional P2 policy route only if operating services require it. Search, thank-you, 404 and error states are utility surfaces outside the marketing route count. /company/about/ redirects to /company/; /company/careers/ redirects to /careers/.

12 Navigation Architecture

Desktop navigation uses two visual levels to accommodate the requested architecture. Main row: logo to Home, Services, Solutions, Industries, Technologies, Work and Resources; primary button “Discuss Your Project” to /contact/. Utility row: How We Work, Company, Careers and Contact. Company dropdown includes Business Information, Engagement Models, Support and Security. A secondary “View Our Work” link appears only if approved work exists; otherwise use “Explore Our Process.”

At narrower laptop widths collapse to a menu before labels wrap. All eleven requested navigation destinations remain accessible in mobile navigation; never place an unavailable destination in the menu. Mega menus have a category heading, two or three short groups and a “View all” link. Click and keyboard activation are supported; hover is an enhancement. Escape closes and restores focus; Tab follows normal document order. Do not assign application-menu ARIA roles to ordinary site navigation.

Footer groups: Company, Services, Solutions, Industries and Technologies, Resources, Legal, Contact. Show full legal name and accurate address label, business email, supported phone, copyright year and verified active social URLs. Use real text links, not icon-only contact methods. On mobile, link groups may be accordions with accessible buttons; identity, contact and legal links stay readily visible.

13 Homepage Specification

The homepage defines 19 modules. Modules with no verified content are omitted entirely; do not publish bracketed placeholders. The prototype may show labelled content slots. Use a 1200px maximum container, 24px desktop gutters and 16px mobile gutters. Unless specified, section padding is 88px desktop, 64px tablet and 48px mobile, with 24px card gaps. Visuals labelled “Illustrative product interface” must not be presented as delivered work.

1 Announcement bar

Copy: “Explore how a software project moves from discovery to support.” Link: “Our process.” Full-width background, centered container, one text row on desktop and wrapping text on mobile. 12px vertical padding. Optional; do not show a fake launch, event or availability notice.

2 Header

Use the navigation specified above. Main row 80px desktop and 64px mobile, with a restrained sticky background. Sticky UI must not hide anchors or keyboard focus. Logo has an accessible brand label. The menu button announces expanded state.

3 Hero

H1: "Software built around your business." Copy: "Turn a product idea or a difficult workflow into a clear development plan. Design, build and maintain web, mobile, SaaS and AI-enabled software around your users, systems and commercial priorities." Publish the capability list only after verification. Primary CTA: "Discuss Your Project." Secondary: "Explore Our Process," replaced by "View Our Work" when work is approved. Supporting line: "Define the scope, delivery responsibilities and next steps before development begins."

Desktop: 7/5-column split, heading limited to approximately 18 characters per line, text width 620px, original static workflow or product interface on the right. Mobile: copy and buttons first, visual beneath; no autoplay video. 112px top and bottom desktop, 56px mobile. The hero image is optimized and not lazy-loaded when it is the LCP candidate.

4 Capability summary

Heading: "From product decisions to production software." Six labelled links: Web Applications; Mobile Products; SaaS Platforms; AI and Automation; Cloud and DevOps; Product Design. Intro: "Choose a starting point based on the problem you need to solve." Desktop six compact items or 3x2 grid; mobile 2x3, switching to one column if text truncates at 320px. Use outline icons plus text, never unlabeled logos.

5 Value proposition

Heading: "Make the important decisions before they become expensive changes." Copy: "Clarify who will use the software, which workflows matter first, and how the system will connect to the tools you already use. A documented plan makes scope, trade-offs and acceptance criteria easier to discuss." Desktop 5/7 split: explanatory text and three short panels titled Scope, Architecture and Delivery. Mobile stacked. CTA: "Plan Product Discovery."

6 Service groups

Heading: "Choose the support your project needs." Five cards: "Strategy and design - Translate user needs into a prioritized product plan"; "Application engineering - Build the interfaces, services and data model"; "AI and integration - Connect systems and introduce evaluated automation"; "Cloud and modernization - Plan releases, infrastructure and system change"; "Quality and support - Test, maintain and extend the software." Desktop 3+2 grid; mobile one column. Cards have one primary link, descriptive labels and no nested clickable controls.

7 Delivery method

Heading: "A clear path from discovery to release." Intro: "Each stage should produce something you can review, from the initial brief to tested software and operating documentation." Show Discovery, Planning, UX/UI, Architecture, Development, Testing, Deployment, Monitoring and Improvement. Desktop 3x3 numbered grid; mobile vertical list. Each stage has one concrete artifact. CTA: "See How We Work." Avoid a nine-node horizontal diagram.

8 Featured projects

Heading: "Work you can examine." Two featured cards on desktop, one on mobile; each includes approved name or anonymized description, industry, problem, approach, stack, duration and verified outcome. CTA: "Read the Case Study." CMS placeholder: [ADD REAL CASE STUDY]. Hide section and Work CTA if none exist. Do not substitute fictional customer screenshots. Images use a consistent 16:10 ratio and accurate alt text.

9 Technology choices

Heading: "Choose technology for the job." Copy: "The right stack depends on the product, your existing systems, the team maintaining it and the cost of operating it. Review the options and trade-offs before committing." Group verified items into Frontend, Backend, Mobile, Data, Cloud, AI and DevOps. Desktop text beside grouped chips, mobile stacked groups. CTA: "Explore Our Technology Approach." Logo approval does not imply vendor partnership.

10 Industries

Heading: "Software shaped by the workflow." Example cards: "SaaS - onboarding, permissions and subscription operations"; "Professional services - client intake, approvals and reporting"; "E-commerce - product, inventory and order workflows." Use only relevant sectors. Desktop three cards, mobile single stack. CTA: "Explore Industry Solutions." Regulated sectors require the section 16 review gate.

11 Engagement models

Heading: "A commercial model that fits the work." Copy: "A defined scope, an evolving backlog and an ongoing support need require different agreements. Compare how scope, billing and changes are handled." Six compact cards: Fixed Scope, Time and Materials, Dedicated Team, Staff Augmentation, Discovery Sprint and Maintenance. Desktop 3x2; mobile stacked. CTA: "Compare Engagement Models." No invented price ranges.

12 Delivery principles

Heading: "Know what you are reviewing and receiving." Four rows: "A written scope - agree what is included"; "Visible progress - review completed work against the backlog"; "A tested release - examine acceptance evidence"; "A planned handover - identify repositories, accounts and operating documentation." These are proposed commitments requiring owner signoff. Desktop 4/8 heading and rows; mobile stacked. CTA: "Explore Delivery Responsibilities."

13 Security and quality

Heading: "Make security part of the engineering plan." Copy: "Discuss access, data handling, dependencies, backups and monitoring alongside features. The controls and evidence required for your project should be defined in its scope." Display verified practices only. Desktop two columns of text and a simple boundary diagram; mobile text first. CTA: "Read Our Security Approach." No certification badges without evidence.

14 Client journey

Heading: "What happens after you get in touch." Four groups contain the ten steps: Inquiry and Discovery Call; Requirements and Proposal; Agreement and Kickoff; Development, QA, Launch and Support. Explain that sending an enquiry creates no development contract. Desktop 2×2 panels; mobile vertical. CTA: "Submit a Project Inquiry." This section describes buying and onboarding, distinct from the engineering method in module 7.

15 Testimonials

CMS slot: [ADD VERIFIED CLIENT TESTIMONIAL]. Require quoted text, permission, name or approved attribution, role, company and date. Prefer a static quotation over an autoplay carousel. Desktop readable 800px text width; mobile full container. Omit the entire section without approved content. Do not show "[VERIFIED TESTIMONIAL REQUIRED]" publicly.

16 Insights

Heading: "Practical notes for your next software decision." Show three to six published articles with category, title, excerpt and accurate reading time. Desktop three columns, mobile one. CTA: "Explore Resources." Hide the module until three useful articles are approved. Publication dates reflect actual publication.

17 Frequently asked questions

Eight questions and proposed answers: "Where should a project start? Start with the users, workflows and constraints; discovery can turn these into a scoped backlog." "Can an existing product be improved? Begin with access, architecture and dependency review before deciding what to replace." "How is cost estimated? Scope, integrations, delivery capacity and risk inform the proposal." "Who owns the software? Ownership, pre-existing components and third-party licences must be set out in the signed agreement." "How are changes handled? Record the request, assess time and cost, and approve changes before work proceeds." "Can you sign an NDA? Raise confidentiality requirements before sharing restricted materials; availability and wording must be agreed." "What happens after launch? The agreement defines handover, any defect period and ongoing maintenance." "How do we start? Send a short project summary and your preferred contact method."

Use native accessible disclosure or tested accordion, 800px width, one or several open items without forced auto-close. All answer text remains accessible in rendered HTML. FAQs help readers; no Google rich-result promise.

18 Final call to action

Heading: "Tell us what you're building." Copy: "Share the problem, the stage of your project and any important constraints. Start with a short summary; confidential files can wait until an appropriate sharing arrangement is in place." CTA: "Discuss Your Project." Single centered 800px block; mobile left-aligned with full-width button. 80px desktop and 48px mobile padding.

19 Footer

Use section 12 footer. Desktop up to six columns in the 1200px container, with a separate identity row. Mobile collapsible navigation plus visible legal/contact links. 64px top and 24px bottom desktop; 40px and 24px mobile. Every visible URL must resolve to approved content.

Homepage SEO: title "Custom Software Development and Product Engineering | [BRAND]".

Description: "Explore custom software, web, mobile and SaaS development, delivery methods and

engagement options. Discuss the scope of your next product.” Remove unsupported service terms before publication.

14 Service Page Specifications

Shared service contract

Each service page contains: breadcrumb; H1 and 50-90-word lead; business problems; buyer fit and poor-fit cases; capabilities; illustrative applications; delivery artifacts; process; verified technologies; architecture decisions; security and data considerations; integrations; schedule drivers; engagement model; example workflow; verified evidence if available; four to six FAQs; related services; final CTA. Target approximately 900-1500 useful words when supported, never pad to meet a word count.

Sample shared copy: “A useful scope describes the users, workflows, integrations and acceptance criteria together. Before development, identify what must be delivered, which decisions remain open and what your team needs to provide.” Timeline copy: “Delivery timing depends on scope, access to existing systems, integration readiness and review turnaround. A proposal should state assumptions, milestones and change procedures.” No universal completion promise.

Each service inherits shared navigation, related process/engagement/security links and the contact destination with a non-sensitive service ID preselected. In the following specifications, H1 is also the default SEO title with “ | [BRAND]”; the meta description is separately supplied. Technology names are examples requiring verification. Illustrative workflows are not case studies.

Custom Software Development

Opening: “Build software around the work your business actually does. Start with the users, decisions and information flows, then define an application that fits your operations and can evolve with them.” Sections: When custom software is justified; Workflow discovery; Application boundaries; Delivery and handover. Specify domain models, role matrices, integrations, data migration and operating documentation. Compare configuration of an existing tool with custom development before recommending a build. Example workflow: enquiry → approval → job assignment → reporting. Architecture decisions: modular monolith versus separate services, data ownership and audit requirements. Security: least privilege and sensitive-field handling. Deliverables: specification, UI, code, tests, deployment guide and agreed training. FAQs: build or buy; changing scope; source-code ownership; support. CTA: “Discuss Your Software Project.” Related: Web Development, API Integration, Technical Consulting. Meta: “Plan custom business software around your workflows, users and integrations. Review delivery stages, ownership and support options.”

Web Application Development

Opening: “Create a browser-based product that lets customers or staff complete real tasks. Plan the interface, permissions, business logic and integrations as one system.” Sections: User journeys; Frontend and backend responsibilities; Integration boundaries; Release readiness. Include authenticated applications, role-based dashboards, customer workspaces and admin tools. Distinguish marketing websites from operational web applications. Architecture: server rendering versus client interaction, session management, caching and API boundaries. Security: CSRF where relevant, authorization on every resource and output encoding. Example: account setup → task submission → staff review → status tracking. Deliverables: responsive application, API contracts, tests and handover. FAQs: mobile browser support; offline needs; existing system integration; hosting ownership. CTA: “Discuss Your Web Application.” Related: UI/UX, API Integration, QA. Meta: “Explore web application development for customer portals, dashboards and business workflows, including architecture, testing and deployment.”

Mobile App Development

Opening: “Design a mobile product around the situations in which people use it. Decide which device features, network conditions and platform requirements matter before selecting an implementation approach.” Sections: Mobile journeys; Platform choice; Device and offline behavior; Store release preparation. Specify iOS/Android scope, push permissions, deep links, accessibility and backend dependencies. Compare native and cross-platform options using actual team capability. Example: capture task offline → synchronize → resolve conflict → notify assignee. Security: secure device storage and token lifecycle. Deliverables: app source, device test matrix, store assets and submission support if contracted. Store approval remains outside supplier control. FAQs: native versus cross-platform; offline behavior; store accounts; updates. CTA: “Plan Your Mobile Product.” Related: Backend/API, UI/UX, Maintenance. Meta: “Plan mobile app development around user journeys, device features, backend integrations, testing and release responsibilities.”

SaaS Development

Opening: “Build a subscription product with a clear model for accounts, organizations, access and billing. Treat onboarding and daily operations as part of the product, not additions after launch.” Sections: Tenant model; Subscription lifecycle; Product administration; Operating costs. Cover invitations, entitlements, billing provider events, plans, support and exports. Architecture: tenant isolation, idempotent webhooks, data partitioning and migration. Example: trial → invite team → select plan → entitlement change → cancellation. Security: tenant-specific access tests and support-user permissions. Deliverables: tenancy ADR, subscription state model, application and operator guide. FAQs: billing provider; data export; tenancy; recurring costs. CTA: “Discuss Your SaaS Product.” Related: Web Development, Cloud, QA. Meta: “Explore SaaS product development, including tenancy, subscriptions, onboarding, integrations and the operational work needed after launch.”

MVP Development

Opening: “Turn the most important product assumption into a focused first release. Define what users need to accomplish, which evidence will guide the next decision and what can wait.” Sections: Hypothesis and audience; Minimum useful scope; Learning plan; Next release decisions. Cover prototype versus production MVP, must-have workflows, validation and technical shortcuts recorded explicitly. Example: sign up → complete core task → provide feedback. Deliverables: prioritized backlog, prototype, scoped release and learning plan. Security fundamentals cannot be deferred merely because the product is an MVP. Timeline drivers include recruitment for testing and stakeholder decisions. FAQs: prototype versus MVP; scope cuts; future rebuilding; success measures. CTA: “Plan Your MVP.” Related: UI/UX, Technical Consulting, SaaS. Meta: “Define and build a focused MVP with clear user journeys, release boundaries, learning goals and a practical path to the next version.”

AI Development

Opening: “Assess where AI can improve a defined task, then test the result against realistic examples. Plan data access, evaluation, human review and operating cost before adding AI to a production workflow.” Sections: Task suitability; Data and evaluation; Human review; Production monitoring. Cover retrieval, classification and assisted drafting only if offered. Define quality thresholds, baseline comparison, latency and cost ceilings. Example: retrieve authorized documents → generate draft → cite evidence → human approves. Security: prompt injection, tenant boundaries, tool permissions and sensitive-data controls. Deliverables: evaluation set, documented limitations, implementation and fallback. Do not imply deterministic accuracy. FAQs: model choice; private data; failure handling; ongoing evaluation. CTA: “Assess Your AI Use Case.” Related: AI Integration, Automation, Cloud. Meta: “Evaluate AI application development with task-specific testing, data controls, human review, monitoring and practical cost boundaries.”

AI Integration

Opening: “Connect AI capabilities to an existing product without losing control of its data or workflow. Define the integration boundary, permissions and fallback before choosing a provider.” Sections: Existing system assessment; Provider abstraction; Data flow; Reliability. Distinguish integration from training a proprietary model. Cover API adapters, quotas, retries, model changes and output validation. Example: inbound request → data minimization → model call → schema validation → user review. Deliverables: integration contract, test cases, usage reporting and operational playbook. Security: no secrets in prompts, restricted tool execution and verified vendor terms. FAQs: provider switching; data use; downtime; cost limits. CTA: “Discuss an AI Integration.” Related: API Integration, AI Development, Maintenance. Meta: “Add AI to existing software with explicit data flows, provider boundaries, validation, cost controls and fallback behavior.”

Automation Development

Opening: “Automate a repeatable process while keeping exceptions visible. Map triggers, decisions and handoffs so the workflow remains understandable when something goes wrong.” Sections: Process mapping; Trigger and action design; Exception queues; Operational ownership. Cover approvals, retries, scheduling, reconciliation and audit records. Example: order received → validation → approval → system updates → reconciliation. Architecture: idempotent steps, durable execution and compensation. Security: scoped integration accounts and approval before consequential actions. Deliverables: workflow map, automation, exception guide and handover. FAQs: what should stay manual; duplicate actions; failed runs; account ownership. CTA: “Review Your Automation Opportunity.” Related: API Integration, Internal Tools, QA. Meta: “Plan business automation with reliable triggers, approvals, exception handling, audit records and clear operational ownership.”

Cloud Development

Opening: “Design application infrastructure around the workload and the team operating it. Make deployment, observability, recovery and cost management part of the architecture.” Sections: Workload needs; Managed services; Reliability decisions; Cost visibility. Cover environments, data stores, queues, object storage and infrastructure configuration. Prefer simpler managed components where they meet requirements. Example: build → deploy to staging → verify → release → monitor. Deliverables: infrastructure definitions, runbooks, cost assumptions and restore procedure. Security: private resources, service identities and secrets. FAQs: cloud choice; account ownership; scaling; recovery targets. CTA: “Discuss Your Cloud Architecture.” Related: DevOps, SaaS, Modernization. Meta: “Explore cloud application development with deployment planning, managed infrastructure, monitoring, recovery and cost-aware architecture.”

DevOps Services

Opening: “Make software releases repeatable and observable. Review how changes move from source control to production, and define the checks and recovery steps needed along the way.” Sections: Delivery pipeline; Environment management; Release controls; Incident readiness. Cover CI/CD, infrastructure as code, dependency scanning, logs, metrics and rollback. Example: pull request → checks → preview → approval → deployment → health verification. Deliverables: pipeline configuration, access map and operating runbook. Security: trusted build sources, least-privileged credentials and protected releases. FAQs: existing pipelines; deployment access; rollback; monitoring ownership. CTA: “Review Your Delivery Pipeline.” Related: Cloud, QA, Maintenance. Meta: “Improve development and release workflows with CI/CD, environment controls, monitoring, infrastructure configuration and rollback procedures.”

API Development and Integration

Opening: “Connect systems through interfaces that are explicit about data, permissions and failure. Define the contract before implementing the exchange.” Sections: API contracts; Authentication; Data mapping; Failure and reconciliation. Include REST or event-driven designs as appropriate, pagination, versioning, webhooks and third-party limits. Example: CRM change → signed event → deduplicated update → reconciliation. Deliverables: OpenAPI contract, integration tests, sandbox setup and ownership map. Security: authorization, signing, replay protection and SSRF defenses for outbound integrations. FAQs: vendor limitations; sync frequency; duplicates; API version changes. CTA: “Discuss Your Integrations.” Related: Web Development, Automation, Backend technologies. Meta: “Plan APIs and integrations with clear data contracts, authentication, versioning, retries and reconciliation between business systems.”

Dedicated Development Teams

Opening: “Add delivery capacity with defined roles, responsibilities and communication. Agree how work is prioritized, reviewed and handed over before a team starts.” Sections: Role requirements; Selection and onboarding; Delivery governance; Continuity. Distinguish managed dedicated team from staff augmentation, where the client directs day-to-day work. Publish only verified skills and actual availability. Deliverables: staffing plan, responsibilities, onboarding checklist and agreed reporting. Security: client-approved access and offboarding. FAQs: management responsibility; time-zone overlap; replacement; billing for absence. CTA: “Discuss Your Team Requirements.” Related: Engagement Models, QA, Technical Consulting. Meta: “Explore dedicated development and staff augmentation models, including role selection, onboarding, communication, billing and continuity.”

UI and UX Design

Opening: “Design the workflows before polishing the screens. Understand users, test key interactions and create interface patterns that development teams can implement consistently.” Sections: Research; Information architecture; Prototyping; Design system and handoff. Cover interviews, task flows, wireframes, content, accessibility and usability findings. Example: research → task map → prototype → test → revision → component specification. Deliverables: design files, tokens, interaction states and annotated acceptance criteria. Security: realistic data handling in prototypes; no production customer records. FAQs: research access; design-only projects; accessibility; developer handoff. CTA: “Plan Product Design.” Related: MVP, Web Development, QA. Meta: “Explore product UI and UX design, from research and workflows to tested prototypes, accessible components and developer handoff.”

QA and Software Testing

Opening: “Make release confidence depend on evidence. Define the important user journeys, failure modes and non-functional requirements, then test them against agreed acceptance criteria.” Sections: Risk-based test plan; Automation; Cross-platform coverage; Release evidence. Cover functional, integration, regression, accessibility and performance testing; specialist penetration testing only if genuinely provided. Example: requirement → test → execution → defect → fix verification → release report. Deliverables: test strategy, cases, automation where appropriate and residual-risk report. FAQs: automation scope; existing codebases; security testing boundaries; release decisions. CTA: “Discuss Software Quality.” Related: DevOps, Web Development, Maintenance. Meta: “Plan software testing around business risk, critical journeys, regression coverage and clear evidence for release decisions.”

Software Maintenance

Opening: “Keep an existing application understandable and operable after release. Define responsibility for defects, dependencies, infrastructure and changes rather than treating every request as the same type of support.” Sections: Takeover assessment; Maintenance scope; Triage; Planned upgrades. Include patching, backups, logs, dependency upgrades and small changes only as contracted. Example: ticket → classification → assessment → approval → fix → verification → release. Deliverables: service inventory, runbook, backlog and support terms. FAQs: inherited applications; defect versus enhancement; coverage hours; third-party outages. CTA: “Discuss Maintenance.” Related: Support, Legacy Modernization, DevOps. Meta: “Explore software maintenance for existing applications, including assessment, defect handling, dependency updates and agreed support coverage.”

Legacy Software Modernization

Opening: “Improve an existing system without assuming everything must be replaced. Identify business-critical behavior, integration dependencies and the safest sequence of change.” Sections: Assessment; Preserve or replace; Data migration; Incremental cutover. Compare refactoring, replatforming and replacement. Example: characterize current behavior → isolate module → test replacement → migrate cohort → monitor → retire old path. Deliverables: assessment, risk register, migration plan and rollback criteria. Security: unsupported dependencies, historical access and data retention. FAQs: rewrite versus improve; downtime; undocumented code; data verification. CTA: “Review Your Existing System.” Related: Cloud, QA, Technical Consulting. Meta: “Plan legacy software modernization through assessment, incremental change, migration verification and documented rollback decisions.”

E-commerce Development

Opening: “Build commerce workflows around products, orders and operations. Decide which platform capabilities to use and where custom development is justified.” Sections: Platform choice; Catalog and checkout; Operations integrations; Release testing. Include inventory, tax-provider integration, fulfilment, returns and subscriptions as scoped. Example: cart → hosted payment → verified webhook → order → fulfilment. Never store raw card data on the marketing website; client-project payment scope needs separate review. Deliverables: storefront or commerce application, integrations and operating guidance. FAQs: platform versus custom; payment provider; inventory sync; ongoing fees. CTA: “Discuss Your Commerce Project.” Related: APIs, UI/UX, QA. Meta: “Explore e-commerce development for catalog, checkout, inventory and fulfilment workflows, with clear integration and operating responsibilities.”

Enterprise Software Development

Opening: “Develop business software within the constraints of your wider technology environment. Plan identity, integration, change control and operational handover alongside the application itself.” Sections: Stakeholder alignment; System boundaries; Governance; Phased adoption. Include SSO, permission models, auditing, procurement and migration dependencies. Example: requirements approval → architecture review → staged delivery → UAT → controlled rollout. Deliverables: governance plan, architecture records, tested software and support handover. FAQs: procurement information; existing standards; access controls; rollout sequencing. CTA: “Discuss Enterprise Delivery.” Related: APIs, Dedicated Teams, Modernization. Meta: “Plan enterprise software with stakeholder governance, identity integration, acceptance testing and controlled operational handover.”

Technical Consulting

Opening: “Get a structured view of a software decision before committing to implementation. Review the context, compare options and document the consequences of each approach.” Sections: Decision scope; Technical assessment; Options and trade-offs; Actionable recommendations. Include discovery, architecture assessment, build-versus-buy and roadmap planning only when offered. Example: questions → access and interviews → assessment → options workshop → written decision. Deliverables: findings, assumptions, prioritized risks and next steps. Security: scoped read-only access and confidential report distribution. FAQs: fixed consulting scope; access required; independent recommendations; implementation follow-on. CTA: “Plan a Technical Review.” Related: MVP, Modernization, Technology Approach. Meta: “Explore technical consulting for software architecture, discovery, modernization and build-versus-buy decisions with documented recommendations.”

15 Solutions Pages

All solution pages use a business workflow hero, current-state pain, proposed system boundaries, user roles, a sample flow, essential modules, integrations, data/security decisions, rollout and adoption plan, related services, FAQs and CTA. H1s below also supply title text followed by “ | [BRAND]”. Each meta description uses the supplied opening condensed to 140-165 characters without a hard truncation rule. SEO editor must preview the actual result. No results or industry experience are implied by illustrative use cases.

Startup Product Development. Opening: “Turn a product idea into a sequence of decisions, prototypes and releases that can be evaluated by real users.” H2s: Identify the first user; Decide the first release; Test the core journey; Plan the next investment. Define founder inputs, discovery deliverables and learning criteria. FAQs: what to validate first; who supplies content. CTA: “Plan Your First Release.” Links: MVP, UI/UX, Engagement Models.

Digital Transformation. Opening: “Start with the business process that needs to change, then plan the systems, data and adoption work around it.” H2s: Map current operations; Prioritize changes; Connect systems; Introduce change in stages. Require current/future process diagrams, dependencies and training owner. FAQs: where to start; how to avoid disruption. CTA: “Map Your Transformation.” Links: Enterprise, Modernization, Consulting.

Business Automation. Opening: “Identify repetitive work that can be standardized and reduce manual handoffs without hiding exceptions.” H2s: Find repeatable tasks; Define rules; Connect departments; Measure process health. Explain orchestration across departments versus the narrower workflow page. FAQs: suitability; exception ownership. CTA: “Review Your Business Process.” Links: Automation Development, Internal Tools, API Integration.

Customer Portals. Opening: “Give customers a clear place to submit requests, review progress and access the information they are authorized to see.” H2s: Customer journeys; Account permissions; Documents and requests; Support handoff. Include invitations, authorization, document access and notifications. FAQs: login options; customer data isolation. CTA: “Plan a Customer Portal.” Links: Web Development, UI/UX, Security.

Internal Business Tools. Opening: “Replace a fragmented operational workflow with a tool built around the people responsible for the work.” H2s: Staff tasks; Role permissions; Data entry and validation; Reporting and training. Include accessibility, bulk actions with confirmation and export permissions. FAQs: spreadsheet migration; administrator ownership. CTA: “Discuss an Internal Tool.” Links: Custom Software, Automation, QA.

CRM Solutions. Opening: “Organize customer relationships around your actual sales and service process, using configuration, integration or custom development where each is justified.” H2s: Pipeline design; Data quality; Integrations; Reporting and adoption. Include deduplication, consent status, activity history and import reconciliation. FAQs: existing CRM reuse; migration. CTA: “Review Your CRM Requirements.” Links: API Integration, Custom Software, Consulting.

ERP Solutions. Opening: “Connect business operations through a defined model for records, approvals and responsibilities.” H2s: Process boundaries; Master data; Module integration; Phased rollout. Distinguish custom modules from replacing a complete ERP. Identify accounting, inventory and permissions dependencies; qualified client experts approve financial rules. FAQs: integration with existing ERP; phased scope. CTA: “Discuss Your ERP Scope.” Links: Enterprise, APIs, Modernization.

SaaS Platforms. Opening: “Plan the complete subscription journey, from onboarding and access to administration and support.” H2s: Account lifecycle; Tenant boundaries; Subscription operations; Product administration. Focus on solution modules; link to SaaS service for implementation detail

rather than duplicating it. FAQs: product operations; plan changes. CTA: “Plan Your SaaS Platform.” Links: SaaS Development, Cloud, Engagement Models.

Marketplace Development. Opening: “Define how buyers, sellers and operators interact before building the marketplace interface.” H2s: Participant roles; Listings and discovery; Transactions and disputes; Operator controls. Document payment-provider boundaries, moderation and dispute ownership. No promise to hold funds or provide financial services. FAQs: payment flows; moderation. CTA: “Discuss Your Marketplace.” Links: Web Development, APIs, E-commerce.

AI Powered Applications. Opening: “Place AI inside a useful workflow with clear limits, review points and a reliable path when the model cannot complete the task.” H2s: Task and evidence; User controls; Evaluation; Operating limits. Require labelled output, feedback and human review for consequential actions. FAQs: incorrect outputs; private information. CTA: “Assess an AI Application.” Links: AI Development, AI Integration, Security.

Analytics Dashboards. Opening: “Bring agreed business metrics into a dashboard whose definitions and data sources are visible.” H2s: Metric definitions; Source systems; Data freshness; Decisions and exports. Explain lineage, permissions, timezone, aggregation and stale-data indicators. FAQs: inconsistent source data; refresh frequency. CTA: “Plan Your Dashboard.” Links: Custom Software, APIs, Databases.

Workflow Automation. Opening: “Make one recurring workflow repeatable, observable and recoverable.” H2s: Trigger and inputs; Rules and approvals; Exceptions; Audit and recovery. Show a concrete illustrative request → approval → action workflow; map retry, duplicate and cancellation behavior. FAQs: failed steps; manual override. CTA: “Map Your Workflow.” Links: Automation Development, API Integration, Maintenance.

Solutions hub H1: “Software solutions for the work you need to do.” Opening: “Explore common systems and workflows, then examine the services and engineering decisions needed to deliver them.” H2s: Build a product; Connect operations; Serve customers; Modernize systems. CTA: “Discuss Your Requirements.” SEO title: “Business Software Solutions | [BRAND]”. Meta: “Explore customer portals, internal tools, SaaS platforms, automation and business systems, with practical delivery and integration considerations.” FAQ: how solutions differ from services; how scope is selected. Links: all published solution pages, Services and How We Work.

16 Industry Pages

Industry pages are P1 unless genuine sector capability makes one central at launch. Use H1 “[Sector] Software Development”, title “[Sector] Software Development | [BRAND]”, and a meta description summarizing the actual workflows listed below. Sections: sector workflows; common system types; integration/data constraints; user roles; delivery considerations; approved relevant work; related services; FAQs; CTA “Discuss Your [Sector] Project”. No claim of past sector clients without evidence.

Sector	Suggested opening copy	Unique content and FAQ directions
Technology	“Plan software that fits your product architecture and the engineering team maintaining it.”	SDK/API contracts, developer workflows, release coordination; ask about existing standards and handover
SaaS	“Design the workflows behind onboarding, permissions and subscription operations.”	Tenant lifecycle, support access and product analytics; ask about isolation and subscriptions

Sector	Suggested opening copy	Unique content and FAQ directions
E-commerce	"Connect storefront experiences with the systems that manage products, orders and fulfillment."	Inventory reconciliation, payment events and returns; ask about platform fit and integration limits
Professional Services	"Organize client intake, project work and approvals around your delivery process."	Client portals, time records, permissions and document handling; ask about confidentiality and workflow migration
FinTech	"Define the software, data and integration boundaries of a financial workflow before implementation."	Ledger boundaries, identity providers, audit records and reconciliation; client compliance team approves regulated flows
Education	"Build learning and administration workflows around the people using them."	Teacher/student roles, content permissions, accessibility and age-appropriate data collection; ask about minors and institutional systems
Healthcare Technology	"Plan healthcare software around authorized access, workflow needs and the sensitivity of the information involved."	Data minimization, access trails, integrations and clinical boundaries; determine applicable obligations before scope approval
Real Estate	"Connect property information, enquiries and transaction administration in a clear operational workflow."	Listing freshness, document permissions and CRM sync; ask about source ownership and updates
Logistics	"Make shipment, dispatch and exception information visible to the people responsible for action."	Carrier APIs, location permissions, offline operations and event reconciliation; ask about stale tracking and connectivity
Travel	"Coordinate booking information and operational updates across customer and supplier systems."	Availability, timezones, cancellation and vendor failures; ask about inventory consistency and refunds ownership
Media	"Organize content workflows, access and distribution around your publishing model."	Rights metadata, moderation, media processing and distribution; ask about licensing and retention
Retail	"Connect store and online operations through consistent product and transaction information."	POS integration, inventory, staff roles and reconciliation; ask about offline needs and stock conflicts
Manufacturing	"Build tools that support production planning, quality records and operational visibility."	ERP/MES boundaries, traceability and disconnected facilities; ask about safety-critical separation and source systems

FinTech, healthcare and education pages must state: "Project requirements, data obligations and any regulatory responsibilities depend on the use case and jurisdictions involved. These must be assessed with your qualified advisers before scope and architecture are agreed." This is proposed neutral scope copy, not an assertion of certification or regulatory approval. Do not advertise clinical decision-making, medical device capabilities or regulated financial operations without appropriate expert review and real delivery capability.

Industries hub H1: "Software in the context of your industry." Opening: "Review the workflows and system constraints relevant to your sector, then discuss the requirements specific to your

organization.” H2s: Customer-facing products; Operational systems; Sector constraints. FAQ: sector experience; regulatory responsibility. CTA: “Discuss Your Requirements.” Meta: “Explore sector-specific software workflows, integrations and delivery considerations across the industries relevant to our services.” Link sectors only when their content is substantive and approved.

17 Technology Pages

Hub H1: “Technology chosen for the product and the team.” Opening: “Architecture decisions should reflect the workload, existing systems, security requirements and the people who will maintain the software. Compare choices in context rather than selecting a stack from a list of logos.” H2s: Selection criteria; Application layers; Operations; Integration boundaries. CTA: “Discuss Your Architecture.” Title: “Technology and Engineering Approach | [BRAND]”. Meta: “Review how software technologies are selected across frontend, backend, data, cloud, automation and integrations.” FAQs: existing stack compatibility; vendor choice; long-term maintenance.

Every category page includes the opening below, selection criteria, appropriate uses, trade-offs, interoperability, testing and maintenance considerations, verified capability records, related service links and two FAQs. Title uses the category H1 followed by brand; meta copy is the short opening plus the relevant selection criteria. No individual vendor keyword pages are required at launch.

H1	Opening copy	Examples to verify and decisions to document
Frontend Engineering	“Choose an interface approach that supports your users, content and interaction requirements.”	React, Next.js, TypeScript; rendering, accessibility, browser support, CMS and analytics; FAQs about SSR and design systems
Backend Engineering	“Design application logic and interfaces around explicit responsibilities and data boundaries.”	Node.js, Python, Java, .NET, PHP/Laravel; throughput, hiring, libraries, API versioning and background tasks; FAQs about monoliths and integrations
Mobile Engineering	“Select a mobile stack around device requirements, release constraints and maintenance.”	Flutter, React Native or verified native capability; offline sync, device APIs, store review; FAQs about shared code and testing
Database Engineering	“Model information around consistency, access patterns and recovery requirements.”	PostgreSQL, MySQL, MongoDB, Redis; transactions, document structure, cache invalidation and retention; FAQs about migration and backup
Cloud Platforms	“Choose cloud services that meet the workload without creating unnecessary operational burden.”	AWS, Azure, Google Cloud; data region, managed services, portability and cost; FAQs about ownership and vendor lock-in
DevOps Tooling	“Make build, test and release behavior repeatable across environments.”	Docker, Kubernetes only where justified, CI platforms; configuration, secrets and observability; FAQs about container need and rollback
AI and ML Engineering	“Select models and supporting systems against a defined evaluation task.”	Verified model/provider capabilities; retrieval, evaluation, data use, latency and human control; FAQs about quality and model updates

H1	Opening copy	Examples to verify and decisions to document
Automation Platforms	"Match workflow tools to the reliability, integration and governance the process requires."	Verify supported orchestration tools; triggers, permissions, retries and human approvals; FAQs about failure handling and portability
Third Party Integrations	"Treat each external service as an explicit dependency with limits and an owner."	Payments, authentication, communication APIs, analytics and CMS; quotas, webhooks, data processing and outage behavior; FAQs about credentials and vendor change

Technology records require status (verified/currently offered/retired), internal capability owner, evidence reference and last review date. Evidence can be a real delivery record or internal capability assessment; it need not be publicly disclosed. Vendor trademarks are descriptive, not partnership claims.

18 Case Study System

Case-study listing H1: "Software projects in context." Opening: "Examine the problem, the decisions and the delivered work behind each approved project." Title: "Software Development Case Studies | [BRAND]". Meta: "Read approved software project stories covering requirements, architecture, delivery and verified outcomes." Filters: service, industry and technology; AND across categories and OR within a category. Default order: editorial featured then publication date. Pagination: 12 per page. Empty filter result: "No projects match these filters. Clear filters or discuss your requirements." No approved work means the public listing and inbound links are withheld, not a fake project gallery.

Detail structure: breadcrumb; verified project title; summary; client name or approved anonymized label; industry; company-size band if permitted; project type; duration; team roles; challenge; business objective; original constraints; discovery findings; solution; architecture; design; stack; development sequence; integrations; testing; security; deployment; obstacles and response; outcome; approved metrics; client feedback; screenshots; related services; CTA "Discuss a Similar Project". FAQ suggestions: applicability to another context; delivery differences; what remains confidential. SEO title: "[PROJECT TITLE] Case Study | [BRAND]"; meta: "[APPROVED PROBLEM AND APPROACH SUMMARY]".

Authoring copy: "The project addressed [VERIFIED PROBLEM] for [APPROVED CLIENT DESCRIPTION]. The agreed scope covered [ACTUAL SCOPE]. The team selected [ACTUAL APPROACH] because [DOCUMENTED REASON]." Outcome: "[INSERT VERIFIED RESULT]". These are editorial placeholders and cannot pass publication validation.

Metrics require value, unit, baseline, measurement window, method, evidence owner and permission. If no numerical evidence exists, describe verified delivered functionality without implying financial benefit. Anonymization requires client permission and removal of identifiable screenshots and metadata. Consent withdrawal unpublishes the story and purges CDN/search caches. Never equate a prototype with a production customer project. Demo work may be separately identified as an "Illustrative internal demonstration" outside the case-study collection.

19 Company Pages

Company page

H1: "A software partner you can understand before you engage." Opening: "[BRAND NAME] provides [VERIFIED SERVICES] through [ACTUAL DELIVERY MODEL]. This page explains who operates the business, how work is organized and where to find company and contact information." Sections: What we build; Mission; Vision; Operating model; Values in practice; Engineering culture; Actual history; Leadership and team; Locations; Business information; Careers.

Proposed mission: "Help organizations turn practical requirements into maintainable software."

Proposed vision: "Make software delivery easier to understand, evaluate and operate." Values copy:

"Make decisions explicit. Document the scope. Review the work. Respect customer information."

Owner approval is required before adopting these as company statements.

Team fields: [ADD ACTUAL TEAM MEMBER], role, approved biography, real photograph or neutral initials, location if approved and genuine profile link. Identify contractors or partners accurately; do not imply all contributors are employees. History and timeline appear only with verified dates and milestones. Show a registered address as registered, not as a staffed office. No stock office photography presented as an actual workplace.

CTA: "Discuss Your Project"; secondary "View Company Information". FAQs: operating locations; who manages work; delivery model. Title: "About [BRAND] | Software Development Company". Meta: "Learn about [BRAND], its verified services, operating model, engineering principles and company contact information."

Business information page

H1: "Company and contact information." Opening: "Find the legal identity and contact details of the business responsible for this website and its services." Fields: legal name, trading name, jurisdiction, registration number where appropriate, registered address, separately labelled operating address, domain, business email, support email and phone. Include last verified date and contact for corrections. Do not publish personal identity documents, private owner residential details or tax documents unnecessarily.

Title: "Business Information | [BRAND]". Meta: "Find verified company identity, jurisdiction and business contact information for [BRAND]." CTA: "Contact the Company." Links: Terms, Privacy, Engagement Models, Support. FAQ: contracting entity; registered versus operating address. Store this information once in CMS settings and reuse it everywhere.

How We Work page

H1: "A defined process from the first discussion to ongoing support." Opening: "Software delivery works best when scope, responsibilities and review points are visible. Each engagement should state what is being built, how decisions are made and what your team receives." Title: "Software Development Process | [BRAND]". Meta: "Review the software delivery process from discovery and design to development, testing, deployment, documentation and support." CTA: "Discuss Your Project." FAQs: cadence; changing priorities; acceptance; delays. Links: Engagement Models, Security, relevant services.

Stages and outputs: consultation establishes fit and restrictions; discovery maps stakeholders and workflows; requirements produce acceptance criteria; specification describes functional and non-functional behavior; wireframes establish layout and journeys; UI design defines components and states; architecture records system boundaries; planning prioritizes the backlog; development

produces reviewed code; QA records test evidence; UAT confirms agreed criteria; security review checks scoped controls; deployment follows a release plan; documentation and training support handover; maintenance and optimization follow the signed agreement.

Proposed cadence: weekly progress update and risk log; demonstrations at agreed sprint intervals; client decisions recorded in the project system. Scrum, Kanban or another method is selected for the work, not claimed simultaneously. Name the business owner, product owner and technical owner. An accepted milestone must identify approver, date and evidence. A change request states description, reason, cost/time effect and approval before implementation.

Engagement Models page

H1: “Choose how the work is scoped, managed and billed.” Opening: “The right engagement depends on how much is known, how quickly priorities may change and who manages day-to-day delivery. Compare the responsibilities and limits before requesting a proposal.” Title: “Software Development Engagement Models | [BRAND]”. Meta: “Compare fixed scope, time and materials, dedicated teams, discovery and maintenance, including billing, changes and project responsibilities.” CTA: “Discuss an Engagement.” FAQs: estimate versus quote; IP ownership; deposits; cancellation; third-party fees. Links: How We Work, Contact, Terms, Support.

Model	Best fit and billing	Advantages and limits	Change control and documents
Fixed scope	Defined outcomes; agreed milestone invoices	Clear scope baseline; new work changes price/timing	Signed statement of work, acceptance checklist, written change orders
Time and materials	Evolving backlog; agreed rates and approved time	Flexible priorities; final cost depends on effort	Timesheets or agreed effort records, budget cap alerts, backlog approvals
Dedicated team	Sustained delivery; monthly capacity agreement	Continuity; capacity does not guarantee fixed output	Role plan, availability, management responsibilities and replacement terms
Staff augmentation	Client-managed delivery; time/capacity billing	Specific skills; client owns prioritization and supervision	Role scope, access rules, interview process, offboarding and time approval
Discovery sprint	Unknown scope; agreed discovery fee	Reduces uncertainty; does not include full build	Agenda, research access, findings, prototype or specification as contracted
Retainer	Ongoing improvement; reserved time or service scope	Predictable access; unused capacity rules matter	Monthly scope, approval limits, rollover rules and reporting
Maintenance	Operating application; agreed coverage fee	Defined support responsibilities; not unlimited enhancement	Service inventory, hours, priorities, exclusions, escalation and renewal terms

Commercial copy: “A proposal identifies the contracting entity, scope, payment schedule, taxes where applicable, third-party costs and acceptance process. Ownership of newly developed work, pre-existing tools and licensed components is defined in the signed agreement. An estimate is not a fixed-price commitment.” Publish actual deposit, refund, cancellation and dispute rules only after contract and counsel review. Client infrastructure and vendor charges should be identifiable; avoid unexplained payments to unrelated entities.

Support page

H1: “Know what support covers after release.” Opening: “Support begins with a shared understanding of the application, its dependencies and the responsibilities in your agreement. Distinguish a defect, an operational incident and a request for new functionality.” Title: “Software Support and Maintenance | [BRAND]”. Meta: “Understand post-launch support, maintenance scope, issue triage, updates, infrastructure responsibilities and escalation.” CTA: “Discuss Maintenance”; existing customers receive “Contact Support” using a verified channel.

Sections: Handover; Defect reporting; Dependency updates; Infrastructure and backups; Planned enhancements; Coverage and escalation. Proposed tiers are descriptive until contracted: Essential maintenance, Managed operations, Continuous improvement. Do not display response or resolution commitments without staffing and contractual support. State [SUPPORTED HOURS AND TIMEZONE], [SUPPORT EMAIL], [ESCALATION CONTACT], [DEFECT PERIOD IF ANY]. Emergency support appears only if actually operated. FAQs: bug versus enhancement; vendor outage; outside-hours request; inherited software. Link Maintenance service, Security and Engagement Models.

20 Contact and Lead Generation

REQ-LEAD-01 P0. H1: “Tell us about your project.” Opening: “Share the problem you want to solve, the stage of your project and any important constraints. A short summary is enough to begin. Please do not send passwords, payment details or confidential customer data.” Title: “Discuss Your Software Project | [BRAND]”. Meta: “Contact [BRAND] to discuss software requirements, project scope, integrations, timelines and engagement options.” CTA: “Submit Project Inquiry.” H2s: Project details; Your contact details; What happens next; Other ways to reach us. FAQs: information needed; confidentiality; response process. Links: Privacy, Engagement Models, How We Work.

Use two logical groups with a clear progress indicator if displayed as steps. Keep required fields from the brief, but provide “Not decided” and “Not applicable” where a prospect cannot reasonably know the answer. Phone, existing-product URL and attachment remain optional. Do not silently turn every unknown into a guessed value.

Field	Required and validation	UI behavior
First and last name	Yes; 1-100 Unicode characters each	Support international names; no Latin-only regex
Business email	Yes; valid syntax, max 254	Accept personal domains for founders; do not require a corporate domain
Phone	No; max 32, normalized if possible	Country prefix helper; no phone required for email contact
Company	Yes; 1-160	Accept “Pre-launch” as an explicit answer
Company website	Yes response; HTTPS URL or “No website yet”	Separate checkbox makes absence valid; never fetch submitted URL server-side
Country	Yes; configured country code	Do not infer citizenship or eligibility
Service needed	Yes; published service ID or “Not sure”	Multi-select capped at 5

Field	Required and validation	UI behavior
Budget	Yes answer; configured range or “Need guidance”	Include currency; ranges editable without code
Expected timeline	Yes; configured choice or “Flexible”	No implied scheduling promise
Project summary	Yes; 30-5000 characters	Counter and examples; preserve line breaks as plain text
Existing product URL	No; URL max 2048	Store as data, never crawl automatically
Attachment	No; up to 3 approved files, 10 MB each	PDF, DOCX, PNG, JPG; visible limit; scanning required
Preferred contact	Yes; email or phone if supplied	Scheduling link is optional, not a forced third-party embed
Privacy acknowledgment	Yes; unchecked	Link notice; no bundled marketing consent

Checkbox copy: “I have read the Privacy Policy and understand how my enquiry will be handled.” If counsel requires consent for this processing, change wording and lawful-basis records accordingly; do not mislabel all processing as consent. Optional newsletter consent, if offered, is separate and unchecked.

State flow: idle → validating → submitting → saved/success or recoverable error. Disable repeated submission while pending; preserve form values on error. Success only follows durable database commit. Copy: “Your enquiry has been received. Reference: [REFERENCE]. We will review the information and contact you using your preferred method.” Follow with three steps: review fit; clarify requirements; agree next discussion. A response target appears only after operational approval.

Alternative contacts: [BUSINESS EMAIL], [BUSINESS PHONE], [ACCURATELY LABELLED ADDRESS]. A blocked bot challenge must offer an accessible alternative contact method. Do not auto-enrol enquiries in marketing or upload their attachments to public media storage.

21 Quote and Estimation Flow

REQ-QUOTE-01 P1. This is a guided enquiry, not an instant price calculator. H1: “Outline the project you want to build.” Opening: “Answer a few questions to prepare a useful project discussion. The final scope, schedule and price require review.” Title: “Request a Software Project Quote | [BRAND]”. Meta: “Outline your software type, stage, features, budget and timing to request a scoped project discussion.” FAQs: whether price is binding; editing answers; confidential requirements. Links: Engagement Models, Privacy, Contact.

Steps: 1 product type (website, web application, mobile, SaaS, AI, automation, existing improvement, other); 2 stage (idea, requirements, design, existing product, modernization); 3 features (authentication, payments, dashboard, API, AI, administration, marketplace, subscriptions, mobile, analytics, other); 4 budget with currency and “Need guidance”; 5 timing; 6 contact details using the contact schema; 7 review and edit; 8 submit and acknowledgement. The eighth step is submission, not an additional questionnaire.

Show “Step X of 7” during input, then a confirmation screen. Back retains answers, including browser back within the flow. Change to a parent answer invalidates incompatible hidden answers explicitly.

Validate only the current step until final review, then validate the complete payload server-side. No automatic price appears. If indicative ranges are added later, version the assumptions and label them non-binding before display.

Keep P0 contact available even if this feature is delayed. Store no PII in localStorage or URLs. In-memory state is the default; refreshing warns about unsaved answers. P1 optional sessionStorage may store only non-personal choices for 24 hours with a visible clear control. Do not store contact details or attachments there. Partial answers are not leads. A unique idempotency key prevents duplicate complete enquiries.

22 Resource Center

H1: “Resources for better software decisions.” Opening: “Explore practical explanations of planning, designing, building and operating software. Use these guides to prepare questions and evaluate trade-offs.” Title: “Software Development Resources | [BRAND]”. Meta: “Read practical software development guides and engineering articles on product planning, architecture, delivery, security and maintenance.” CTA: “Discuss Your Project.” H2s: Planning a product; Engineering decisions; Operating software; Glossary and FAQs. FAQ: who writes resources; how articles are reviewed. Links: relevant services and technology categories.

Blog listing H1: “Software engineering and product insights.” Opening: “Practical notes on decisions that affect how software is built and maintained.” Title: “Software Engineering Blog | [BRAND]”. Meta: “Explore articles on software development, SaaS, AI, cloud, testing and product design.” Filter by the 13 requested categories; show only categories with published posts. Search matches title, summary and body, excludes drafts, and preserves filter state in non-sensitive query parameters. Twelve results per page; no infinite-scroll-only access.

Article fields: title, slug, summary, author reference, reviewer when used, category, tags, body blocks, hero, alt text, source links, published and updated dates, related services, SEO and review due date. Article H1 is the actual title; opening states the decision the article helps with. Layout: summary, jump links, explanation, examples, trade-offs, checklist, sources, related content and contextual CTA. No invented author or credential; organization attribution is allowed where accurate. FAQs only when useful. Individual meta description is written from the article’s actual contents.

Initial editorial backlog: How to write a software project brief; Fixed scope versus time and materials; What an MVP should include; How to plan an API integration; What to ask about software ownership; Preparing for a legacy system assessment; Choosing between custom software and an existing tool; A practical software handover checklist; How to evaluate an AI feature; Understanding software maintenance costs; What a customer portal needs; Planning accessible product interfaces. These are proposed topics, not already published articles. Release at least three reviewed pieces before showing the homepage Insights module; then target two substantive pieces per month subject to editorial capacity.

23 Careers

H1: “Work on software with clear expectations.” Opening: “Learn how work is organized at [BRAND] and review the roles currently open.” Company-specific culture, benefits, employment type, remote policy and locations require verification. Title: “Careers at [BRAND]”. Meta: “Explore verified openings at [BRAND], the hiring process and expectations for working with the team.” CTA: “View Open Roles.” Secondary contact appears only if recruiting enquiries are supported. H2s: Working here; Engineering principles; Hiring process; Open roles; Internships. FAQs: remote eligibility; interview stages; applications without an opening.

No openings state: “There are no open roles listed at the moment.” Do not collect speculative CVs unless an approved process and retention policy exist. Benefits must not be decorative promises.

Job template: H1 “[ROLE]”; opening states real mission and team context; sections Responsibilities, Required skills, Helpful experience, Employment type, Location/remote eligibility, Compensation where required/approved, Hiring stages, Accessibility adjustments, Apply. Title: “[ROLE] Careers | [BRAND]”; meta derived from real role and location. Fields: job ID, status, hiring owner, location, valid-through date, employment type, responsibilities, requirements and application endpoint. CTA: “Apply for This Role.” Application fields: name, email, job ID, CV, optional portfolio URL and role-specific questions; no unnecessary demographic fields. Record privacy notice version. Closed roles disable applications and remove JobPosting markup; show closure and link other openings. Existing applications are retained according to approved policy.

24 Trust and Security

Trust hub H1: “Information for evaluating how we work.” Opening: “Find company information, policies and the security practices relevant to our services. Project-specific requirements should be confirmed during scoping.” Title: “Trust and Company Policies | [BRAND]”. Meta: “Find company information, privacy, security and data-handling resources for evaluating [BRAND].” H2s: Business identity; Security; Data handling; Policies; Questions. CTA: “Ask a Security Question.” FAQ: documentation availability; certifications; project-specific requirements. Link Business Information, Privacy, Security, Data Processing, Terms and Support. Do not expose confidential audits or internal infrastructure details publicly.

Security page H1: “Security practices and project responsibilities.” Opening: “Security requirements depend on the application, its users and the data involved. This page describes verified company practices and how project-specific requirements are agreed.” Title: “Security Practices | [BRAND]”. Meta: “Review verified security practices, project responsibilities, data handling and the process for reporting a security concern.” CTA: “Contact the Security Team” only if an actual monitored contact exists; otherwise “Contact Us About Security”.

Sections: secure development lifecycle; source control and review; authentication and authorization; encryption and secrets; infrastructure and environments; backup and recovery; logging and monitoring; dependencies and vulnerabilities; employee/customer access; third-party services; data minimization; continuity; incident response; responsible disclosure. Each claim includes an internal evidence owner, last verified date and scope. Website controls required in section 44 are build requirements; they are not proof the company applies the same controls to all client projects.

Sample conditional copy: “Access to [VERIFIED SYSTEMS] is limited to authorized roles and reviewed [ACTUAL CADENCE].” “Backups cover [ACTUAL DATA] with [ACTUAL RETENTION]; restoration is tested [ACTUAL CADENCE].” These placeholders must be resolved before publication or replaced with narrower approved statements. No SOC 2, ISO 27001, HIPAA or PCI DSS claims without appropriate

evidence and scope.

Disclosure copy: “If you believe you have found a security issue affecting this website, contact [SECURITY EMAIL] with a description and the affected URL. Please do not include unnecessary personal data.” Counsel and security owner must approve scope, safe-harbor language if any and response expectations. Do not invent a bug bounty or public availability history.

25 Legal and Compliance

Legal content must match the real entity, markets, processors, business model and contracts. The following is a content specification, not a ready-to-publish legal opinion. Appoint qualified counsel for applicable jurisdictions before publication. Avoid importing unrelated policy text from competitors.

Page and H1	Proposed introductory copy	Required sections and publication inputs
Privacy Policy	“This notice explains how [LEGAL NAME] handles personal information collected through this website and project enquiries.”	Controller identity, data categories, purposes, applicable legal bases, recipients, transfers, retention, rights, contact, notice version and effective date
Website Terms	“These terms describe the conditions for using this website. Software project services are governed by a separate signed agreement.”	Operator, permitted use, website IP, third-party links, disclaimers, liability subject to law, dispute route, applicable law approved by counsel
Cookie Policy	“This page explains the storage and tracking technologies used on this website and how to change your preferences.”	Actual inventory, provider, purpose, category, duration, first/third party, controls and withdrawal
Accessibility Statement	“We aim to make this website usable by people with different access needs and provide a way to report barriers.”	Evaluated scope, target, testing date, known limitations, contact and remediation; no unsupported conformance claim
Data Processing Information	“Project data-handling responsibilities are agreed according to the services provided and the parties’ roles.”	Controller/processor roles, DPA availability where relevant, approved subprocessors, data regions, deletion/return, incident cooperation and contact

Each policy title is “[PAGE NAME] | [BRAND]”; meta description briefly states that policy’s purpose. H2s follow the required sections above. CTA: “Contact Us About This Policy.” FAQ suggestions: scope, contacting the responsible team and version changes; do not add speculative legal answers. Cross-link company identity, relevant related policies and contact. Policy text must be readable without accepting optional cookies.

The Engagement Models page contains non-binding summaries of contracts, IP ownership, third-party licences, payment schedules, refund/cancellation rules and escalation. If public purchasing is introduced, reassess dedicated service terms, refunds, consumer rights, tax and AUP requirements. No universal “no refunds” statement or blanket claim of legal compliance.

26 Content Strategy

REQ-CONTENT-01 P0. All content follows draft → factual review → editorial review → approval → scheduled/published → review due → archived. Business-specific claims reference an evidence record. A reviewer distinct from the author approves claims when staffing permits; restricted fields always require publisher authority.

Publishing gate: required content complete; service actually offered; company identity reconciled; permissions attached for logos, people and customer material; facts approved; links resolve; metadata and alt text complete; no bracket placeholders, lorem ipsum, dummy counters or stale job data. Regex detection is a helper, not the whole review: detect ADD, VERIFIED, INSERT and common dummy content in customer-visible fields. Block placeholder publication rather than merely hiding text in CSS. Also exclude blocked records from APIs, search and structured data.

Write explanatory paragraphs and decision-oriented examples. Use original diagrams labelled illustrative when appropriate. Do not portray stock images as staff, offices or delivered products. Exact company contact details come from one settings record. Review contact and legal fields monthly, service capability quarterly and technical articles every six months or after material product changes. Review dates are internal workflow dates; do not change an article's public updated date without a real edit.

Hub copy standards: Services H1 "Software development services from planning to support"; opening "Explore the work involved in defining, building and operating a software product. Choose a service to review scope, deliverables and typical decisions." H2s are the five service groups. CTA "Discuss Your Project". FAQs address choosing services and combining engagements. Title "Software Development Services | [BRAND]"; meta "Explore custom software, product design, application development, integration, testing and support services with clear delivery information." Links point to published individual service pages and Engagement Models.

27 SEO Strategy

Map intent before assigning keywords. Commercial service intent belongs to Services; system intent to Solutions; sector workflow intent to Industries; implementation evaluation to Technologies; informational questions to Resources. Do not generate all service × industry × city permutations. Search volumes and competitive ranking difficulty have not been measured here and must not be fabricated.

Intent cluster	Primary destination	Supporting content and links
Custom software development company	Custom Software service	Build-versus-buy guide → service → enquiry
SaaS development services	SaaS service	Tenant model guide and SaaS solution → service
MVP development	MVP service	Scope and product brief guides → discovery CTA
API integration services	API service	Webhook/reconciliation articles → service
Legacy modernization	Modernization service	Assessment checklist → consulting and modernization

Intent cluster	Primary destination	Supporting content and links
Customer portal development	Customer Portals solution	Permissions guide → web application service
Industry software	Relevant approved industry page	Workflow article and approved case → appropriate service
React or cloud architecture	Technology category	Technical comparison → relevant delivery service

Each commercial page links its parent, two or three genuinely related services/solutions and a relevant article or approved case. Articles link to a useful service only where relevant. Avoid exact-match keyword repetition. H1 describes the page; H2s reflect questions and decisions. Page titles and meta descriptions are editorially unique; snippet lengths are guidelines, not ranking guarantees.

Current research check on 4 October 2026: Google’s official documentation update log states FAQ rich results stopped appearing from 7 May 2026 and the documentation was removed in June. Keep visible FAQ content for users; omit FAQPage as an SEO deliverable unless a separate semantic need is justified. Recheck implementation guidance before release. Sources and research limitations appear in section 28.

28 Design Direction

Use an editorial technology-company aesthetic: a mostly light canvas, strong dark typography, an occasional dark engineering section, precise diagrams and asymmetric layouts that support reading order. Avoid large animated particle backgrounds, logo carpets without context, automatic carousels and decorative code that does not mean anything.

Current competitive research

Research date: 4 October 2026. The following official sites were reviewed through their retrievable page content and navigation. This supports information architecture, content depth and CTA observations. It is not a rendered mobile, measured typography, visual pixel or performance audit. The type scale, spacing and mobile behavior in this PRD are original recommendations, to be validated in design and browser QA.

Reference	Observed pattern	Selected application
thoughtbot homepage	Services grouped into development, design, product and team/process areas; prominent contact path and case/resource access	Group capabilities around decisions and work types; keep one clear enquiry CTA
thoughtbot Distribute case study	Case narrative connects early product work with discovery and design	Explain what was learned and why choices were made, rather than showing screenshots alone
Netguru services	Broad offer divided into specific service families and linked detail pages	Use a navigable service taxonomy with substantial leaf pages

Reference	Observed pattern	Selected application
Netguru bespoke development	Business-use-case explanations and a content outline support a deep service page	Use anchored page contents and scope-specific explanations for long pages
Thoughtworks client stories	Industry and capability links connect portfolio exploration with relevant work	Use meaningful case filters and contextual cross-links, with real evidence only

These references illustrate patterns, not an endorsement ranking or permission to reuse their copy, marks, photos or results. Do not borrow their claims about scale, experience or customers.

Research references

S1 thoughtbot homepage - <https://thoughtbot.com/> - navigation, service grouping and contact paths.

S2 thoughtbot Distribute case study - <https://thoughtbot.com/case-studies/distribute> - narrative case structure.

S3 Netguru services - <https://www.netguru.com/services> - service categorization.

S4 Netguru bespoke development - <https://www.netguru.com/services/ bespoke-software-development> - deep commercial page structure.

S5 Thoughtworks clients - <https://www.thoughtworks.com/clients> - industry/capability discovery.

S6 W3C WCAG 2.2 - <https://www.w3.org/TR/WCAG22/> - accessibility reference for section 32.

S7 Google Web Vitals - <https://web.dev/articles/vitals> - field-performance thresholds in section 46.

S8 Google Search documentation updates - <https://developers.google.com/search/updates> - FAQ feature retirement and current SEO verification.

S9 Next.js Server and Client Components - <https://nextjs.org/docs/app/getting-started/server-and-client-components> - component boundary guidance.

S10 OWASP File Upload Cheat Sheet - https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html - upload risk controls.

S11 ICO Cookies and similar technologies - <https://ico.org.uk/for-organisations/direct-marketing-and-privacy-and-electronic-communications/guide-to-pecr/cookies-and-similar-technologies/> - UK-specific privacy reference, not a substitute for jurisdiction review.

All references accessed 4 October 2026. Numerical product targets, retention defaults and deployment decisions in this PRD are proposed requirements, not competitor benchmarks or universal legal requirements.

29 Design System

Token group	Specification
Brand colors	Primary #1D4ED8; hover #1E40AF; dark anchor #0F172A; restrained secondary teal #0F766E
Surfaces and text	Canvas #FFFFFF; alternate #F8F AFC; text #0F172A; secondary #475569; decorative border #CBD5E1; input boundary #64748B
Semantic states	Success #166534 on #F0FDF4; warning #92400E on #FFFBE B; error #B91C1C on #FEF2F2; information #1E40AF on #EFF6FF; always include text/icon
Typography	Inter if licensed/self-hosted; system sans fallback. Code uses a licensed monospace. Body 18/28 desktop and 16/26 mobile; small 14/22; avoid body smaller than 16
Headings	H1 56/64 desktop, 36/42 mobile; H2 40/48 and 28/36; H3 24/32 and 22/30; weights 600-700; test actual wrapping
Spacing	4, 8, 12, 16, 24, 32, 48, 64, 88, 112px; text paragraphs 16px; section heading to content 32px
Layout	Max width 1200px; reading width 720px; desktop 12-column grid with 24px gap; cards 24-32px padding
Shape	Buttons/inputs 8px radius; cards 12px; chips fully rounded only when short; no excessive pills
Elevation	Default flat border; raised hover shadow 0 8px 24px rgba(15,23,42,.08); dialogs stronger scrim and shadow
Controls	Minimum designed target 44×44px; buttons 48px high; inputs 48-52px; textarea at least 144px
Focus	Visible 2px high-contrast outline with 3px offset; test against each actual surface; never remove without replacement
Motion	120-180ms for hover/focus, up to 240ms disclosure; transform/opacity only when useful; reduced-motion removes nonessential transitions

Buttons: primary filled, secondary outline, tertiary text, destructive explicit label. Define default, hover, focus, pressed, disabled and loading. Disabled contrast may be muted but label remains readable; loading shows text and progress, never an unlabeled spinner. Inputs define empty, filled, focus, invalid, disabled and read-only. Error text sits next to the field and is linked by aria-describedby.

Cards vary by content type; avoid identical decorative cards for every paragraph. Navigation uses clear active state. Modal uses labelled dialog, focus trap, Escape, inert background and focus return. Accordion uses button and expanded state. Tabs use keyboard navigation only when implementing genuine tab panels; prefer links for navigation. Tooltips cannot contain essential instructions. Tables retain headers and a labelled scroll region when necessary. Pagination exposes current page and real links. Skeletons preserve dimensions without implying successful data loading. Toasts supplement durable in-page messages, never replace submission confirmation.

30 Component Library

Component family	Required inputs and states	Acceptance focus
Header, mega menu, mobile menu	Published navigation tree, active path, CTA; open/closed	Keyboard, focus return, no dead links
Announcement, breadcrumb, footer	Approved text, link targets, canonical hierarchy, identity settings	Hidden if empty; consistent identity
Hero, split section, CTA banner	H1, lead, media, CTA labels/targets	One H1, mobile order, no text baked into images
Service, solution, industry, technology cards	Record reference, title, summary, optional icon	One primary card link, readable focus
Case, team, article cards	Verified record, image/alt, metadata	Suppress unapproved records everywhere
Testimonial, logo cloud, statistics	Claim evidence and publication permission	Cannot render with missing evidence
Process timeline and journey	Ordered steps, descriptions, optional artifacts	Logical reading order; no horizontal trap
FAQ, tabs, modal, tooltip	Accessible labels, content, state	Keyboard and screen-reader behavior
Forms and uploads	Schema, labels, options, validation, errors	Server validation, retry, privacy and scanning
Search, filters, pagination	Query, allowed facets, counts, page	Accessible state, no draft leakage, URL persistence
Cookie controls	Categories, consent version, current choice	Equal access to reject/manage, scripts blocked correctly
Empty, loading, success, error	Context-specific message and recovery action	No blank surface, stack trace or false success

Maintain a component catalogue with variants, spacing, props, keyboard behavior, screenshots and test stories. CMS blocks use an allowlisted component map. No arbitrary HTML, scripts, custom styles or unreviewed embeds from the marketing editor.

31 Responsive Design

Viewport	Required behavior
320px	16px gutters; single-column text/forms; wrap long email/URLs; no clipped CTAs; visual headings may reduce to 32px
375px	Standard compact mobile; cards stack; buttons full-width when two labels cannot fit
390px	Verify common phone layout, sticky header offsets and software keyboard behavior
430px	Preserve mobile navigation; avoid stretching line lengths or small two-column forms
768px tablet	24px gutters; two-column cards where content fits; forms remain primarily one column
1024px laptop	Collapse crowded nav; two/three-column grids; no label wrapping in navigation
1280-1440px desktop	Full navigation when it fits; 1200px container, 12-column editorial layouts
1920px and wider	Center capped container; background may extend; do not stretch paragraphs across screen

Test 200% text zoom and reflow at 320 CSS pixels. Avoid fixed card heights. Forms use input types/autocomplete for mobile keyboards. Sticky CTA is optional and must not obscure errors, keyboard focus, cookie controls or content; default is no sticky bottom CTA. Horizontal scrolling is limited to genuine wide data/code, with clear labeling and keyboard access. Do not hide substantive content on mobile to reduce page height.

32 Accessibility

REQ-A11Y-01 P0 targets WCAG 2.2 AA using the W3C reference S6. Provide landmarks, skip link, one logical H1, ordered headings, visible labels, meaningful link names, appropriate image alternatives and keyboard access. Normal text contrast target is at least 4.5:1; large text 3:1; meaningful UI boundaries 3:1. AA target-size minimum is 24×24 CSS pixels with specified exceptions; this project chooses 44×44 for primary controls as a stronger usability target.

Focus must not be obscured by sticky elements. Avoid drag-only interactions. Authentication must support password managers and paste. Forms identify errors in text, associate messages with inputs and move focus to an error summary after invalid submit. Dynamic result counts and submission status use restrained live announcements. Respect reduced motion. Provide captions and appropriate alternatives for meaningful media. Do not use an accessibility overlay as a substitute for implementation.

QA must combine automated scans with keyboard and screen-reader journeys: navigation, consent, form validation, successful enquiry, filtering and job application. Record tested combinations and unresolved limitations. Claim conformance publicly only after a scoped assessment; the launch target itself is not a certification.

33 Functional Requirements

ID	Priority	Requirement and verification
REQ-NAV-01	P0	Every visible navigation item resolves to a published page or valid anchor; crawl and keyboard test
REQ-CONTENT-01	P0	CMS review and evidence gate controls publication; invalid record cannot publish
REQ-LEAD-01	P0	Valid enquiries persist once before success and enter the operational queue
REQ-UPLOAD-01	P0 if uploads enabled	Private quarantined upload, scanning, authorization and expiry; otherwise remove attachment UI
REQ-EMAIL-01	P0	Confirmation and internal notification processed independently with retry and delivery status
REQ-CRM-01	P0 queue, P1 external CRM	Durable lead handling survives CRM downtime without duplication
REQ-SEO-01	P0	Published canonical pages produce server-rendered content, metadata and valid sitemap entries
REQ-A11Y-01	P0	Core journeys meet scoped WCAG 2.2 AA acceptance assessment
REQ-PRIV-01	P0	Privacy notice, preference controls, retention and deletion operate as specified
REQ-SEC-01	P0	Authorization, rate limiting, secrets, backups and monitoring pass security tests
REQ-CMS-01	P0	Editors can update standard content without code and cannot grant their own privileges
REQ-WORK-01	P0 template, conditional content	Only approved actual case studies appear in page, API, feed and search
REQ-JOBS-01	P0 conditional	Genuine openings accept applications; closed jobs reject new applications
REQ-QUOTE-01	P1	Guided enquiry retains step state and submits an idempotent complete record
REQ-SEARCH-01	P1	Search and filters return only published, permitted content
REQ-NEWS-01	P1	Newsletter double opt-in and unsubscribe; no implicit subscription

ID	Priority	Requirement and verification
REQ-PORTAL-01	P2	Separately scoped authenticated client portal with tenant isolation

MVP is the complete public-site platform and approved content, not all future business functions. No custom chat system, booking engine, billing system or public checkout is necessary at launch. If Careers is visible without vacancies it provides the truthful empty state. Case-study and resource modules with insufficient evidence remain unpublished.

P2 portal scope: client login, project overview, milestones, deliverables, contracts, invoices, documents, messages, tickets, status, meeting notes, files, approvals and support requests. Use a separate application boundary and authenticated routes. Every object requires tenant and project membership checks; never rely on unguessable IDs. Add immutable approval records, file scanning, audit events, notification preferences and retention policies. Reuse a suitable existing platform before commissioning this subsystem. It needs its own threat model, schema and acceptance plan before development.

34 CMS

Use a managed headless CMS with structured content and an approved component library. Procurement must verify roles, revisions, draft preview, auditability, export, data region and webhook support before selecting a vendor. Next.js is the presentation layer; the CMS is authoritative for marketing content; the operational PostgreSQL database is authoritative for enquiries and applications. Do not maintain two editable copies of the same content.

Content types: Page, Service, Solution, Industry, TechnologyCategory, Technology, CaseStudy, Post, Author, TeamMember, Job, FAQ, Testimonial, Navigation, SiteSettings, SEO and Evidence. Each record has stable ID, slug where routed, status, locale, author/editor, created/updated/published timestamps, revision, reviewer and review due date. Content relationships use IDs, not duplicated titles. Evidence contains claim, source reference, owner, verification date, permission status and expiry if applicable; sensitive evidence remains restricted.

P0 workflow: save draft → preview through signed short-lived access → review → publisher approval → publish webhook → invalidate affected page, hub, sitemap and search index → verify live version. Publish and unpublish webhooks are signed and replay-protected. Draft mode is authenticated, no-store and noindex. Public APIs never expose draft data. Delete a referenced item only after resolving incoming links or creating an appropriate redirect.

Block library: text, image, heading, cards, process, FAQ, evidence quotation, verified statistics, comparison table and CTA. Allow rich text with a controlled element set; disallow script, iframe and arbitrary event attributes. Media includes rights, source, alt text, dimensions and focal point. CMS marks missing alt text unless the asset is explicitly decorative.

35 Admin Dashboard

Prefer the CMS's own content interface plus a small protected lead console. Avoid recreating a full CMS. If the chosen CRM already provides the needed secure lead workflow, it may replace the custom lead console, but local durable intake and reconciliation remain required.

Role	Allowed operations	Explicit restrictions
Editor	Draft pages, posts, metadata and media	No publish, identity/legal edits or lead access by default
Publisher	Review and publish approved content	Cannot bypass required evidence or change user roles
Sales	Read assigned/all leads as approved, add notes, change status	No careers data, content publishing or credentials
Recruiter	Read assigned jobs/applications, update hiring status	No sales exports or candidate marketing enrollment
Operations admin	Manage integrations, roles and system settings	Sensitive actions require reauthentication and audit
Auditor	Read approved audit views	No mutation and no unnecessary attachment access

Dashboard modules: enquiry totals/status, integration failures, content awaiting review and jobs needing closure. Leads show filters, assignment, status history and audit trail. Content management includes services, cases, posts, team, jobs, media, SEO, navigation and settings through CMS. Audit records include actor, action, target, timestamp and safe change summary; do not log full enquiry payloads or secrets. Export is separately permissioned, logged and protected against CSV formula injection. Bulk destructive actions require explicit confirmation and recoverable soft deletion where appropriate.

Authentication uses managed identity with MFA and invite-only staff accounts. No public staff signup. Session expiry and revocation must take effect on server actions, not only UI. If self-hosted passwords are unavoidable, use a current supported password-hashing library and tuned Argon2id configuration, recovery protections and no plaintext password storage.

36 CRM and Lead Flow

The server accepts a validated enquiry, writes submission and lead plus outbox events in one database transaction, then returns success. Background workers send confirmation, create internal notifications and synchronize CRM independently. Email or CRM failure cannot erase a saved lead or cause the browser to submit a second lead.

Flow: visitor → validation and abuse controls → durable intake transaction → response; outbox independently fans out to confirmation, internal alert and CRM sync. A reconciliation job compares queued/sent/failed states and retries recoverable failures. Dashboard shows failed tasks with reference IDs and safe retry actions. Internal alerts contain a short summary and protected lead link, not attachments or full sensitive narratives.

Lead statuses: New → Contacted → Qualified → Proposal → Won or Lost. New/Contacted/Qualified may move to Spam after review. Reopen Lost to Contacted with reason. Record every transition, actor and timestamp. Automation never marks Won based solely on a form submission. Lead assignment uses configured service ownership or a fallback inbox; unassigned leads appear in a visible queue.

Use the submission UUID as external CRM idempotency identifier. A later genuine enquiry from the same email is a new submission, optionally related to an existing lead after review. Do not merge solely on shared company domain. Source fields are allowlisted UTM values and a clean landing path;

strip arbitrary URL query strings, email addresses and tokens. Preserve an unknown-source state when consent or blockers prevent attribution.

Internal target proposals: review new non-spam leads each business day; escalate unassigned records older than one business day. These are operating recommendations, not public response guarantees until staffing is confirmed.

37 Email Architecture

Use a transactional provider with verified sending domain, SPF, DKIM and an appropriate monitored DMARC rollout. Keep transactional and marketing streams separate. Use consistent legal sender identity, reply-to address and support link. Branded HTML must be readable with images blocked; provide equivalent plain text. Escape all user data. Do not include confidential summaries in email by default.

Template	Subject	Suggested body and action
Contact confirmation	Your project enquiry has been received	"Hello [FIRST NAME], thank you for contacting [BRAND]. Your reference is [REFERENCE]. We will review your enquiry and contact you using your preferred method. Reply to this email if you need to correct a detail." Link process; no unsupported SLA
Quote confirmation	Your project outline has been received	"Thank you for sharing your project outline. Reference [REFERENCE]. The next step is to review the scope and clarify any open questions. This acknowledgement is not a project quote or agreement."
Internal new lead	New project enquiry [REFERENCE]	"Service: [SAFE SERVICE]. Source: [SAFE SOURCE]. Assigned to: [OWNER/UNASSIGNED]. Review this enquiry in the secure dashboard." Protected link only
Career confirmation	Application received for [ROLE]	"Thank you for applying for [ROLE] at [BRAND]. Your reference is [REFERENCE]. The hiring team will review your application according to the role's process." No promise of interview
Newsletter verification	Confirm your subscription to [BRAND]	"Confirm that you want to receive [ACTUAL CONTENT TYPE]. If you did not request this, ignore this email." One-use signed token, 24-hour expiry; no subscription before confirmation
Support acknowledgement	Support request received [REFERENCE]	"We have received your request. Its priority and next steps will be reviewed under your support agreement. For the agreed escalation route, see [SUPPORT LINK]."

Provider events are signed webhooks. Track queued, sent, delivered, bounced and failed; delivery does not prove reading. Retry transient failures at 1, 5, 30 and 120 minutes then daily until 72 hours, with a capped attempt count and dead-letter queue. Permanent bounces stop retries and surface to

the owner. A contact form must not become an email-bombing relay: rate limit confirmations and repeated recipient addresses, use fixed templates and never echo arbitrary links.

38 Analytics

REQ-ANALYTICS-01 P0. Use one approved analytics platform; tag manager is optional and access-controlled. Disable session replay and keystroke capture by default. Do not send names, email, phone, summaries, file names, candidate information, raw URLs or full IPs as custom analytics properties. Operational lead records and anonymous analytics have different purposes and retention.

Events	Trigger and allowed properties
page_view	On consented initial view and route transition once; page_id, page_type, clean path
service_view, case_study_view, blog_view, job_view	Once per relevant view; public content ID and category
cta_click	Intentional activation; cta_id, placement, destination_type, page_id
contact_started	First field interaction once per form session; form_version
contact_submitted	After durable accepted response; random event_id for deduplication, form_version; no lead PII
quote_started, quote_step_completed	First step and valid completed step; step_id, flow_version; no free text
quote_submitted	After accepted response; event_id, flow_version
phone_click, email_click, calendar_click	Click only; channel and placement, not actual address/number
newsletter_subscribed	After verified opt-in; event_id, content preference category if allowed
job_application_started, job_application_submitted	Start and accepted response; public job_id only

Consent-eligible common fields: source, medium, campaign, landing page, device category, coarse country only where appropriate and privacy-reviewed. Allowlist lengths and characters; reject unexpected properties. Do not persist marketing identifiers when consent is refused. Country need not be inferred for a useful launch dashboard.

Dashboards: landing page → commercial page → enquiry start → accepted enquiry; quote step abandonment; source quality based on separately reviewed CRM aggregates; content-assisted journeys within consent limits; job funnel; API failure and delivery metrics from operational monitoring. A CTA click is not a lead. Use server counts for actual accepted submissions and annotate differences from consented analytics. Newsletter conversion counts confirmation, not merely the request.

39 Integrations

Integration	Priority and boundary	Failure behavior
CMS	P0; server read token and signed webhook	Serve previously published cached content; alert stale content
Transactional email	P0; server-only credentials	Durable outbox retry; enquiry remains saved
Object storage and scanner	P0 if uploads enabled	Quarantine; disable file access until clean; accept enquiry without file if user chooses
CRM	P1 or P0 when existing operation needs it	Retry/deduplicate; local inbox remains source of intake truth
Analytics	P0 approved minimum	Consent-gated; never block site or form
Search Console	P0 setup	Ownership verification and indexing reporting; no user data transfer from forms
Scheduling	P1	External link or consent-aware embed; show email fallback
Monitoring	P0	Scrub PII before export; alert delivery tested
Bot prevention	P0 adaptive controls	Accessible fallback; do not trust client token without verification
Internal Slack notification	P1	Limited metadata and protected link; authorized workspace only
Newsletter	P1	Separate opt-in, unsubscribe and suppression synchronization

Maintain vendor inventory: owner, purpose, transferred fields, processing region, contract/DPA status, retention, secret location, webhook validation and outage runbook. No secret is placed in NEXT_PUBLIC variables or CMS public content. Environment files are not committed. The website may link to real LinkedIn, GitHub, X, YouTube or Instagram accounts; hide missing or inactive placeholder URLs.

40 Frontend Architecture

Use current stable, mutually compatible Next.js App Router, React and TypeScript versions selected and locked during implementation. Tailwind implements design tokens; shadcn/ui primitives may be adapted and tested rather than assumed accessible. Motion library use is optional and limited to meaningful interactions. No version number in this PRD should be treated as a dependency lock.

Server Components render content, metadata and CMS reads; small Client Components handle menus, consent, forms, filters and interactive state, consistent with official Next.js guidance S9. Use static rendering/revalidation for marketing content and dynamic server endpoints for intake. Keep authenticated and preview responses no-store. Avoid turning the root layout into a client component.

Suggested structure: app/(marketing) for public routes; app/(admin) for protected operations; app/api for route handlers; components/ui for primitives; components/blocks for editorial sections; lib/content

for CMS adapters; lib/validation for shared schemas; lib/auth for permission checks; lib/analytics for consent-aware events; lib/server for database and integrations. Enforce server-only boundaries in imports.

Route metadata reads published records and produces title, description, canonical and social preview. Cache tags include record ID and affected hub IDs. Unpublishing must invalidate details, references and sitemap; preview cookies cannot contaminate public cache. Public forms have progressive enhancement where feasible; core reading/navigation works without client JavaScript. Unknown routes call genuine not-found handling.

41 Backend Architecture

Use Next.js route handlers as a modular backend for intake, authorization, CMS webhooks and read APIs where required. Do not introduce NestJS and microservices merely to support a corporate website. A separate worker/job service processes durable outbox tasks because short-lived web requests must not assume background execution continues after response.

Use managed PostgreSQL for operational records, a supported ORM/migration tool such as Prisma if selected, private object storage and a transactional email provider. The marketing site does not need its own copy of CMS data tables. API schemas, business services and repository adapters remain separate so provider changes are manageable.

Write sequence for intake: parse bounded body; verify origin/bot token as applicable; rate-limit; validate normalized values; verify clean upload tokens; transactionally insert submission, lead and outbox; commit; respond. Workers lock jobs safely, record attempts and idempotency, perform side effects and persist results. Failed jobs remain visible. Database unavailable means return a retryable error, not a success screen or an email-only lead.

42 Database Architecture

Operational database

Use UUID primary keys, timestamptz UTC timestamps and foreign keys. Store enums through checked values or reference tables. Encrypt at rest through the managed platform; application-level encryption may be added for selected fields after key-management review. Avoid storing request bodies in logs. Soft deletion is not sufficient for privacy erasure; the purge process removes data according to the approved schedule.

Table	Important fields and relationships
users	id, identity_provider_subject unique, email, display_name, active, last_login_at; no password when external identity used
roles, permissions	id, key unique, description; user_roles and role_permissions are many-to-many joins
leads	id, owner_user_id nullable FK, status, company, normalized_email, source_summary, created_at, updated_at, closed_at; status indexed
contact_submissions	id, lead_id FK, first_name, last_name, email, phone nullable, company, website nullable, no_website boolean, country, service_ids JSON array, budget_code, currency, timeline_code, summary, preferred_contact, notice_version, acknowledgment_at, source fields, created_at

Table	Important fields and relationships
quote_requests	id, lead_id FK, submission_id optional FK, flow_version, product_type, stage, features JSON array, budget_code, currency, timeline_code, contact fields or FK to contact record, final_summary, notice_version, submitted_at; no abandoned contact drafts
lead_status_history	id, lead_id FK, from_status, to_status, actor_user_id, reason, changed_at
lead_notes	id, lead_id FK, author_user_id FK, body_plaintext, created_at; access restricted to sales
upload_objects	id, submission/application association nullable until claim, object_key unique, original_name restricted, mime, byte_size, checksum, scan_status, expires_at, created_at; purpose enum enquiry/CV
applications	id, cms_job_id, job_version, name, email, portfolio_url nullable, resume_upload_id FK, answers JSON, status, assigned_recruiter_id, notice_version, submitted_at
newsletter_subscribers	id, normalized_email unique, state pending/active/unsubscribed, token_hash, token_expires_at, confirmed_at, unsubscribed_at, consent_version; marketing-only purpose
consent_receipts	id, anonymous_receipt_id, policy_version, categories JSON, captured_at, expires_at; store no contact identity unless needed and approved
outbox_events	id, aggregate_type, aggregate_id, kind, payload_ref, status, attempt_count, next_attempt_at, locked_until, provider_id, last_error_code; ready-job composite index
idempotency_keys	scope, key unique within scope, request_hash, result_reference, response_code, created_at, expires_at; 24-hour default
integration_events	id, provider, external_event_id unique per provider, event_type, received_at, processed_at, processing_status; bounded payload retention
audit_logs	id, actor_id, action, entity_type, entity_id, safe_diff JSON, request_id, created_at; append-only application permissions
settings	key unique, value JSON, version, updated_by; operational settings only, not duplicate editable corporate identity
retention_jobs	id, entity_type, entity_id, due_at, legal_hold boolean, purge_status, completed_at; restricted legal hold changes

Relationships: users receive roles through user_roles; leads have many submissions, notes, status events and outbox tasks; submissions reference zero or more private upload objects; applications reference a published CMS job ID and a CV object; each outbox event refers to its aggregate without embedding unrestricted PII. Referential checks prevent attachment reassignment across anonymous upload sessions or staff scopes.

Deduplication: unique idempotency key for retries, not unique lead email. Index normalized email for authorized search; add created_at/status/owner indexes for queues. Do not index free-text summaries broadly. Use transaction boundaries around submission + lead + outbox writes. Migration scripts use expand/contract changes and verify rollback compatibility.

CMS model and optional custom CMS mapping

CMS entities include services, solutions, industries, technologies, case_studies, posts, authors, categories, tags, faqs, testimonials, team_members, jobs, media, seo_metadata and navigation. Each routed record references SEO and structured body blocks. Case studies relate many-to-many to technologies, services and industries. Posts relate to author, categories, tags and related commercial records. Jobs have publication status, expiry and hiring-owner metadata; applications remain outside the public CMS.

If custom CMS is approved later, implement corresponding tables and join tables such as case_study_technologies, post_tags, post_categories and record_faqs. Include content_versions, publication_reviews, claim_evidence, redirects and media_rights. Only one system may edit each record. A separate analytics_events table is optional for self-hosted consented analytics, with event_id, name, page_id, allowlisted properties and expiry; do not copy analytics events into the enquiries schema by default.

43 API Design

All endpoints require HTTPS, bounded payloads, strict schemas and consistent error envelopes. Public mutation requests accept JSON only except controlled upload transfer. Unknown fields are rejected or explicitly stripped. No endpoint accepts an arbitrary external URL to fetch. Rate limits below are proposed starting settings, tuned for shared networks and abuse telemetry; they are not security guarantees.

Endpoint	Contract	Access and initial limit
POST /api/contact	Contact schema, clean upload tokens, notice version, idempotency key; returns reference	Public; 5 attempts per 15 min/IP plus recipient and global abuse controls
POST /api/quote	Versioned complete quote schema; returns reference	Public; 5/15 min/IP, same deduplication semantics
POST /api/uploads/init	Purpose, MIME, bytes, checksum; returns short-lived scoped upload token	Public abuse-controlled; 3 files/session, 10 MB each; 10/hour/IP
POST /api/uploads/complete	Owned token and checksum; returns pending/clean/rejected status	Token-bound; 30/min/session; never makes object public
GET /api/uploads/{token}/status	Scan state only, no file bytes	Session/token-bound; polling backoff, max 30/min
POST /api/newsletter	Email and explicit preference; generic 202 to prevent enumeration	P1; 3/hour/recipient plus IP controls
POST /api/newsletter/confirm	One-use token	P1; token hash/expiry checked; idempotent
POST /api/newsletter/unsubscribe	Signed token or authenticated preference flow	P1; easy opt-out; no login required with valid token
GET /api/services	Published list; optional allowed category/cursor	60/min/IP, cached; omit if frontend reads CMS server-side only

Endpoint	Contract	Access and initial limit
GET /api/case-studies	Published filters and cursor, max 24 results	Cached; no unpublished evidence or client-private fields
GET /api/blog	Published category/tag/query/cursor	Cached; bounded query length and no expensive wildcard search
GET /api/jobs	Active public jobs only	Cached with expiry-aware invalidation
POST /api/jobs/{id}/apply	Active job check, candidate schema, clean CV token	5/hour/IP and job/email abuse controls; closed job returns 409
POST /api/support	Request and contact reference	Only if support intake enabled; verify authorization for private project details
POST /api/webhooks/cms	Signed event ID, timestamp, record ID and action	Signature plus replay protection; queues invalidation
POST /api/webhooks/email	Signed delivery event	Provider verification and unique external event ID
GET /api/admin/leads	Paginated authorized lead list	Staff authentication, sales permission, no-store
PATCH /api/admin/leads/{id}	Allowed status/assignment with version	Permission and optimistic concurrency; audit required
GET /api/admin/uploads/{id}/download	Authorized clean object access	Short-lived signed download after scope check; no public caching

Success example: {"ok":true,"data":{"reference":"ENQ-[OPAQUE ID]","status":"received"},"requestId":"[UUID]"}.

Validation error: {"ok":false,"error":{"code":"VALIDATION_ERROR","message":"Please check the highlighted fields."},"fields":{"email":"Enter a valid email address."},"requestId":"[UUID]"}.

Opaque references must not encode email or company name.

Status codes: 201 new durable record; 200 identical idempotent replay; 202 newsletter request or queued scan; 400 malformed payload; 401 unauthenticated; 403 unauthorized; 404 unknown public record; 409 conflicting idempotency payload, version conflict or closed job; 413 oversized payload; 415 unsupported type; 422 valid JSON with invalid field values; 429 limit with Retry-After; 503 temporary storage/dependency failure preventing intake. Do not expose stack traces or provider secrets. Same key and same request returns original success; same key and different payload returns 409. Expired key allows a new request but UI should not silently resend later.

44 Security Requirements

REQ-SEC-01 P0. Maintain a threat model covering spam/email bombing, injection, CSRF, XSS, credential theft, IDOR, malicious uploads, leaked drafts, preview-cache contamination, forged webhooks, exfiltration through analytics and unauthorized exports. Appoint control owners; verify implementation rather than displaying a security badge.

Transport and browser controls: HTTPS redirect; secure TLS from managed provider; HSTS after domain/subdomain readiness review; Content-Security-Policy with tested explicit sources,

frame-ancestors and report-only rollout before enforcement; X-Content-Type-Options nosniff; appropriate Referrer-Policy and Permissions-Policy. Do not use broad unsafe-eval or arbitrary script origins for convenience. Separate public caching from personalized responses.

Validation and session controls: server-side schemas and length limits; parameterized database access; safe rendering/sanitization of rich text; authorization for every staff action; CSRF token/origin defenses for cookie-authenticated mutations; strict CORS where applicable; HttpOnly, Secure and appropriate SameSite cookies; rotate sessions after login, revoke on offboarding, rate-limit authentication. Public forms also validate allowed origin and abuse signals, while recognizing that origin headers alone do not stop non-browser bots.

File controls based on OWASP S10: extension/MIME/signature checks, randomized object keys, size/count limits, private quarantine, malware scanning and authorization on access. Reject executable, archive and macro-enabled formats. DOCX is allowed only with appropriate inspection/scanning; never execute embedded content. Decode/re-encode images and strip metadata where practical. Serve approved downloads as attachments from an isolated storage origin. Scan outage keeps file quarantined; the visitor can submit without it. Expire unclaimed files after 24 hours. Use short-lived download links, default 5 minutes, and log authorized downloads without content.

Secrets and operations: managed secret store, environment separation, least-privilege service identities, no production data in previews, encrypted backups, tested restores, dependency and secret scanning in CI. Security patches follow a documented triage process; propose urgent critical-risk triage within one working day, with deployment timing based on exposure and test results. This is an internal target unless actually staffed and contracted. Incident playbook names owner, containment steps, investigation records, communication decision maker and legal review of any notification obligation. Never invent a universal breach-notification deadline.

45 Privacy Requirements

REQ-PRIV-01 P0. Maintain a data map from browser through application, database, email, CRM, file storage and monitoring. Establish applicable privacy rules with counsel using actual markets and entity details. The cookie reference S11 is UK-specific; do not apply it as a universal legal conclusion.

Product default: necessary-only operation until optional tracking choices are saved. Provide equally discoverable Accept all, Reject non-essential and Manage preferences. Categories: Necessary always on, Analytics off, Marketing off. Explain purposes plainly; no prechecked optional categories. Persist a versioned receipt in a first-party necessary preference mechanism. Withdrawal is available from the footer, blocks future optional calls and clears relevant first-party identifiers where feasible. Tag loading and embedded media must respect the choice, not just the visual banner.

Proposed retention schedule, subject to owner/counsel approval: enquiries with no ongoing relationship 12 months after last meaningful activity; unsuccessful applications 6 months after closure unless lawful separate agreement; quarantined/unclaimed uploads 24 hours; operational request logs 30 days; audit metadata 12 months; analytics 6 months; pending newsletter tokens 24 hours and unconfirmed records 7 days; subscribed records until withdrawal plus minimal lawful suppression evidence. Contract/client records follow a separately approved business retention schedule. Record legal holds explicitly and restrict who can apply them.

Deletion workflow: verify requester proportionately → identify relevant records → check applicable retention/hold → delete or anonymize live data and vendor copies → log minimal completion evidence → ensure restoration does not revive erased data. Backups expire on schedule; after restore replay the deletion ledger before resuming operations. Do not promise immediate physical deletion from immutable backups. Candidate data is never used for sales marketing. Staff access and vendor

exports are limited to purpose.

46 Performance Requirements

Field targets from Google Web Vitals S7: 75th-percentile LCP ≤ 2.5 seconds, INP ≤ 200 milliseconds and CLS ≤ 0.1 , evaluated separately for mobile and desktop where sample size is adequate. A new website may lack sufficient field data; prelaunch lab tests are a proxy, not proof of field compliance.

Proposed budgets: key marketing page initial compressed JavaScript ≤ 180 KB excluding explicitly activated third-party tools; initial page transfer ≤ 1.2 MB on representative pages; hero image ideally ≤ 250 KB at common desktop size and ≤ 120 KB mobile; no autoplay hero video; API intake p95 ≤ 1.5 seconds excluding file scanning under expected load; cached page TTFB target ≤ 800 ms in target regions. Measure the actual build and document exceptions with impact and owner.

Use responsive AVIF/WebP with fallback, explicit image dimensions, lazy load below-fold media, preload only critical resources, self-host limited font weights, subset fonts where licensing permits, avoid render-blocking external scripts and split interactive code by route/component. Cache published CMS responses with explicit invalidation. Compress responses and serve static assets through CDN. Lighthouse mobile performance target ≥ 90 is a useful regression signal under a recorded test configuration; it does not replace CWV or accessibility assessment.

47 SEO Technical Requirements

REQ-SEO-01 P0. Use lowercase descriptive hyphenated routes with one trailing-slash policy, one canonical HTTPS hostname and permanent redirects for alternate forms. Each indexable page has a 200 response, unique metadata, one descriptive H1, crawlable links and absolute self-canonical. Canonicalize tracking parameters to the clean page; do not canonicalize unique pagination pages all to page one.

Generate XML sitemap from published indexable records only, using actual material-change lastmod dates. Exclude drafts, previews, admin, search results, thank-you pages and private files. Robots.txt points to the sitemap. Noindex is not access control; protect private surfaces with authentication. Do not block crawling of a URL that must be seen for its noindex to work. Stage/preview environments require access protection and noindex as a second layer; production deployment tests must reject accidental global noindex.

Structured data: Organization on the real entity with verified URLs/contact details; BreadcrumbList on canonical hierarchy; Article/BlogPosting for articles; Person only for real approved people; JobPosting only for genuine active roles and current eligibility; case studies may use Article/CreativeWork semantics matching visible content. ProfessionalService only if semantically appropriate after validating the type and actual business representation. Do not add self-serving review stars or unsupported AggregateRating. FAQPage is not a Google rich-result objective under the checked current guidance.

Provide Open Graph and X card title, description and correctly sized approved image, with no private preview URLs. Alt text describes image purpose, not keyword lists. Listing pagination uses real links and stable order. Filter/search results default noindex when they create duplicate combinations; valuable curated landing pages get their own approved content. Broken routes return 404; temporarily failed resources return appropriate error statuses rather than soft-404 200 pages. Redirect only to a relevant replacement, never blanket-redirect every removed page to Home. Maintain redirects in a reviewed registry and avoid chains.

48 DevOps

Use Git with protected main and short-lived branches. Pull requests require review, type checking, linting, targeted unit/integration tests, security/dependency checks and build verification. Changes to schema, privacy, auth or intake require explicit tests for affected risks. Do not write tests that merely assert static copy equals itself.

Environments: local development with synthetic data; authenticated previews per PR using non-production secrets; staging with production-like configuration and synthetic fixtures; production. Never send staging emails to actual prospects. Use provider sandbox or recipient allowlist. Test integrations against sandboxes or controlled accounts.

CI/CD builds an immutable versioned artifact, runs migrations through an authorized job, deploys to staging, executes smoke tests, then releases with an accountable approver. Store build SHA, dependency lockfile and migration version. Rollback application code independently where compatible; use expand/contract migrations to avoid destructive coupling. Back up before risky data operations. Document emergency rollback and rehearse it before launch.

49 Deployment

Recommended baseline: managed Next.js hosting such as Vercel, managed PostgreSQL in an approved nearby region, managed headless CMS, private S3-compatible object storage, a durable job service, transactional email and error/uptime monitoring. This is a product architecture recommendation, not a price comparison. Procurement must validate current cost, regional availability, runtime limits and contracts before selection. Cloudflare or AWS can substitute only after checking framework/runtime compatibility and operational ownership.

Deployment responsibilities: business owns domain and billing accounts; engineering manages scoped access; operations owns uptime and recovery; marketing owns approved content. Configure apex/www canonical routing, DNS verification, certificate renewal, sender DNS, storage policies and environment-specific endpoints. Record domains and vendor account ownership in handover. Do not place all production services in a developer's personal account.

Release procedure: freeze reviewed content → backup → deploy immutable build → run migrations if compatible → warm critical cached routes → validate forms, policy links, email queue and monitoring → enable indexing only after checks → observe error rate and rollback if critical paths fail. Site publication itself is outside this PRD deliverable; the development team executes it under its release process.

50 Monitoring

Monitor public availability every minute from at least two regions; critical page response/content; API failure rate; database connections/storage; outbox oldest age; email bounces; webhook failures; worker health; scan queue; certificate expiry; dependency alerts and content publish failures. Synthetic tests must use a clearly tagged test enquiry and never pollute sales metrics.

Proposed alert thresholds: two consecutive multi-region availability failures; intake 5xx >2% over 10 minutes with minimum 20 requests; any failed synthetic intake; oldest pending outbox >15 minutes; repeated scan failures; database capacity >80%; certificate expiry under 14 days if not automatically resolved. Low-traffic sites also use absolute failure alerts because percentages alone are unreliable.

Alert routing has a primary and backup owner and an acknowledgement/escalation procedure. Error monitoring scrubs form fields, tokens and file names before export. Record request ID, release version and safe endpoint name. Uptime or SLA claims must not be published merely because monitoring exists.

51 Backup and Recovery

Proposed internal targets: operational database RPO ≤ 1 hour and RTO ≤ 4 hours, contingent on the selected platform and staffing. Enable point-in-time recovery if supported and test the targets. Daily snapshots alone do not satisfy a one-hour RPO. If the vendor cannot meet the target, revise architecture or approve a clearly documented target change before launch.

Daily encrypted snapshots retained 30 days; object versioning and recovery retention aligned with privacy policy; CMS revisions plus scheduled content exports; infrastructure and configuration stored in version control without secrets. Backup credentials are separate from application write credentials. Restrict deletion of backups and alert failures. Backups should survive a single application credential compromise.

Restore monthly into an isolated environment: recover database to a chosen time, restore required media, reconcile CMS references, verify row counts and representative records, replay deletion records, test intake and permissions, measure elapsed time and document result. During real recovery, reconcile provider/CRM side effects using idempotency IDs before re-running jobs. Runbook states who declares an incident and who authorizes reopening production.

52 Testing Strategy

Area	Required evidence
Functional	Contact/quote/job happy paths and invalid input, duplicate retry, back/refresh, outage and closed-job cases
CMS	Draft isolation, role restrictions, publication gates, unpublish invalidation, redirects and rollback to a content revision
Responsive	Screens at all section 31 widths, long text, missing media, software keyboard and zoom
Accessibility	Automated scan plus keyboard, NVDA/Firefox or equivalent, VoiceOver/Safari; critical journey notes
Performance	Recorded mobile lab setup, representative routes, image/JS budgets and postlaunch field plan
Security	Permission boundary, IDOR, CSRF, stored/reflected XSS, injection, webhook replay, secret scanning and upload tests
SEO	Crawl, canonicals, statuses, metadata, headings, sitemap, robots, schema, pagination and redirects
Integrations	Email retry/bounce, CRM duplicate prevention, scanner outage, CMS outage and token rotation
Privacy	Reject-all network inspection, category selection, withdrawal, no PII in analytics/logs, retention/deletion rehearsal
Recovery	Database/media restore with measured recovery targets and deletion replay

Automate high-risk validation, authorization, idempotency and rendering tests. Use end-to-end tests for core journeys. Manual review remains necessary for content truth, visual layout, screen-reader usefulness and policy scope. Test environment fixtures must be clearly synthetic and not publishable. Release gate: no open critical or high-risk defects affecting data security, intake, accessibility of core journeys or false public claims; accepted lower-risk issues have owner and due date.

53 Browser Compatibility

Support current and previous major releases of Chrome, Edge, Firefox and Safari at implementation freeze, plus current and previous supported Mobile Safari/iOS and Chrome Android combinations feasible for the target audience. Record exact versions and devices in QA; “current” must not remain an untested assumption. Include a real iPhone/Safari and Android/Chrome check, not only desktop emulation.

Core content must remain readable when optional scripts are blocked. Test browser autofill, password managers for admin, date/input differences, focus behavior and file upload. Older unsupported browsers receive graceful content and a contact alternative, not a blank page. Define any minimum OS support after audience evidence and platform support review.

54 Wireframes

Homepage wireframes are embedded section by section in section 13. Shared page shell: full-width header; 1200px main container; breadcrumb above title; 720px maximum long-form reading width; final CTA and footer. Each layout uses semantic reading order first. Designer supplies desktop 1440, tablet 768 and mobile 390 designs plus 320px stress checks and interactive states.

Service page

Desktop: breadcrumb → 7/5 hero text and meaningful visual → 8/4 body and sticky anchor list → alternating full-width capability/process blocks → evidence → FAQ → CTA. Anchor list contains Problems, Scope, Delivery, Architecture, Engagement and Questions; its width is about 260px. Mobile: hero stacks, contents becomes a disclosure above body, capabilities become single-column cards. Spacing 80px between major groups, 32px inside groups, 48px section spacing mobile. Primary CTA appears in hero, after scope and at end; no sticky obstructive sales overlay.

Case study

Desktop: title/summary full width; factual metadata strip; large approved screenshot; 8/4 narrative and project facts; architecture figure with caption; outcome and evidence; related services. Mobile: metadata becomes definition list and narrative precedes secondary facts. 16:10 screenshots, text alternatives for diagrams, no tiny text embedded in images. CTA “Discuss a Similar Project” after the outcome. Hide absent metrics/quotation modules without leaving space.

About page

Desktop: intro 7/5 with real photo only if available; mission/operating model two columns; values as short editorial rows; verified timeline; real team grid; labelled locations; legal information link; careers CTA. Mobile: one column, timeline linear, team cards 1-2 columns based on fit. Company facts are text, not decorative counters. 80px group spacing desktop and 48px mobile.

Contact page

Desktop: 8-column form and 4-column next steps/contact sidebar. Labels above controls; first/last name may share a row, other fields full width or logical pairs. Mobile: one column; explanatory intro, form, then alternative contact. Error summary immediately above form, successful receipt replaces form with reference and next steps. Keep legal notice next to submission. 24px between field groups and 8px label spacing. Never place a large map before the form.

Blog listing

Desktop: H1 and intro → category links/search → featured article if editorially selected → three-column cards → pagination. Mobile: stacked search and category disclosure, one-column cards, explicit page links. Count and empty state appear after filters. Keep search/filter state in URL and maintain focus after update. Card hierarchy: category, title, excerpt, real author/date.

Blog article

Desktop: category/breadcrumb → title and summary → real author/reviewer/date → optional hero → 720px body with 240px contents sidebar when space allows → sources → related articles/service CTA. Mobile: contents disclosure above body; code/tables have labelled overflow; headings wrap. Body 18/30 desktop, 16/26 mobile. No sales popup interrupts reading.

Careers listing

Desktop: introduction and work principles → verified policies/benefits → hiring process → role filters and list rows. Mobile: all stacked; job rows show title, location and employment type without truncation. Empty state replaces only role list; no fake roles or “Apply” button without a real destination. FAQ and accommodation contact remain visible.

Job page

Desktop: role header and metadata → 8-column description plus 4-column application summary; full application below or linked to same-page anchor. Mobile: job text then application. Closed banner replaces application controls immediately; keyboard focus must not land on removed input. CV upload shows progress, scanning and rejection states separately.

Security page

Desktop: narrow introductory text → anchor list beside factual practice sections → policy links → security contact. No decorative shields that imply certification. Mobile: contents disclosure and readable full-width sections. Include last reviewed date only when accurate. Sensitive evidence is requested through contact, not downloadable from public cards.

55 User Stories

Story ID	User need	Requirement and completion evidence
US01	As a prospect, I want to understand services so I can judge fit	Service content and navigation; AC01, AC02
US02	As a CTO, I want architecture and practices so I can assess compatibility	Technologies, Security, How We Work; AC03
US03	As a founder, I want commercial options so I can plan an engagement	Engagement Models and honest estimate boundaries; AC04
US04	As a reviewer, I want consistent company identity and policies	Single-source identity; AC05
US05	As a visitor, I want an accessible way to make an enquiry	REQ-LEAD-01; AC06-AC10
US06	As a returning visitor, I want submission retries not to create duplicates	Intake idempotency; AC08
US07	As a privacy-conscious visitor, I want to reject optional tracking	REQ-PRIV-01; AC15-AC16
US08	As a buyer, I want real project evidence with context	REQ-WORK-01; AC02, AC18
US09	As a job seeker, I want only genuine current roles	REQ-JOBS-01; AC12
US10	As a candidate, I want my CV private and acknowledged	Uploads/application flow; AC11-AC13
US11	As an editor, I want reusable content blocks without code access	REQ-CMS-01; AC19
US12	As a publisher, I want unverified claims blocked	REQ-CONTENT-01; AC02
US13	As a sales owner, I want every accepted enquiry recoverable during an outage	REQ-CRM-01; AC09
US14	As a recruiter, I want role-specific access without unrelated sales data	RBAC; AC14
US15	As a reader, I want to search and filter published resources	REQ-SEARCH-01; AC20
US16	As a mobile visitor, I want forms and menus usable at small widths	REQ-A11Y-01 and responsive; AC17
US17	As an operator, I want an alert when intake fails	Monitoring; AC23
US18	As an owner, I want a verified recovery procedure	Backup requirements; AC24

Story ID	User need	Requirement and completion evidence
US19	As a subscriber, I want to confirm and later stop newsletters	REQ-NEWS-01; AC25
US20	As an SEO editor, I want accurate indexable URLs and metadata	REQ-SEO-01; AC21-AC22
US21	As a buyer, I want to understand support and software ownership	Engagement/Support content; AC04
US22	As a security reporter, I want a real monitored contact	Security page and escalation test; AC23
US23	As an administrator, I want exports and role changes audited	Admin requirements; AC14
US24	As a quote requester, I want to review answers before submitting	REQ-QUOTE-01; AC10

56 Acceptance Criteria

AC01 Navigation. Given published destinations, when a visitor uses desktop or mobile navigation by keyboard, then every displayed link works, the menu can be dismissed and focus returns predictably. With no approved work, the Work link and associated CTA are omitted or replaced by Process, not routed to an empty gallery.

AC02 Publication truth. Given a record containing an unresolved placeholder, unapproved claim or missing customer permission, when publication is attempted, then the CMS blocks it and explains the missing approval. It cannot appear in public HTML, API, sitemap, search or schema. Given permission withdrawal, unpublishing removes all references and cached public versions within the configured invalidation target of five minutes.

AC03 Technical evaluation. Given a CTO evaluation task, when the visitor opens a service and follows relevant links, then architecture decisions, testing, deployment and maintenance boundaries are available in no more than three navigation choices. The page names only verified capabilities.

AC04 Commercial clarity. Given a proposed engagement, when the visitor reads Engagement Models, then billing basis, scope handling, change approval, ownership boundaries and support references are explicit, and no unapproved price or binding promise appears.

AC05 Company consistency. Given a verified legal name/address/contact change, when settings are published, then footer, contact, company information and structured data use the new approved value, and policy references are reviewed for consistency. Registered and operating addresses retain correct labels.

AC06 Valid enquiry. Given valid required fields and acknowledgment, when Submit is activated, then one durable submission, related lead and outbox jobs are committed before a receipt is shown. The receipt contains a non-sensitive reference and next steps.

AC07 Invalid enquiry. Given missing or invalid values, when Submit is activated, then no lead is saved, focus reaches an accessible error summary, each error links to its field and valid entries remain present. Errors never require color recognition alone.

AC08 Duplicate and timeout. Given a committed enquiry whose response was lost, when the same key and payload are retried within 24 hours, then the original receipt is returned with no additional lead or email jobs. Reusing the key with different content returns 409 and a recoverable message.

AC09 Dependency failure. Given email or CRM outage, when an enquiry is committed, then success remains truthful, jobs retry and operations sees queue state. Given database failure before commit, then no success is shown and the form supports safe retry.

AC10 Guided quote. Given a partially completed quote, when Back or Edit is used, then answers remain accurate; final review exposes all choices and incompatible hidden values are removed. Submission uses the full schema and returns no automatic binding price.

AC11 Upload security. Given an oversized, disguised, infected or unscanned file, when upload/attachment is attempted, then the file cannot become downloadable or attached as clean. A clean file is accessible only to authorized staff with a short-lived URL. Another session cannot claim its token.

AC12 Jobs. Given an active job, a valid application is accepted. Given a job closed during completion, when Submit is used, then the server rejects new applications with a clear closure message, preserves an optional non-sensitive recovery path and does not misstate acceptance. Expired jobs lose JobPosting markup.

AC13 Application privacy. Given a candidate CV, when a sales-only user attempts access, then access is denied server-side. Confirmation email contains no CV attachment. Deletion removes candidate records and storage according to the approved retention process.

AC14 Staff authorization. Given any staff role, when a forbidden route or object ID is requested directly, then access fails regardless of hidden UI. MFA, revoked sessions, role changes and exports are tested. Sensitive actions produce audit records without full PII payloads.

AC15 Consent refusal. Given a new visitor, when no optional choice has been made or Reject non-essential is selected, then optional analytics/marketing scripts and events do not load or transmit. Core reading and enquiry submission still work.

AC16 Consent withdrawal. Given Analytics permission, when the visitor withdraws it, then future analytics stops and relevant local identifiers are cleared where feasible. The receipt records the updated preference without storing contact identity unnecessarily.

AC17 Mobile and accessibility. Given 320px width, 200% zoom, keyboard use or supported screen reader, when completing navigation and enquiry journeys, then no essential content/control is clipped or obscured, labels/errors are announced and focus order is logical. Intentional code/data overflow is labelled.

AC18 Evidence and case filters. Given approved cases, when filters change, then results follow documented AND/OR logic and a clear empty result appears when appropriate. Every displayed metric has verification and permission; no fake clients fill an empty state.

AC19 CMS operations. Given an editor, when content is drafted, previewed and submitted, then it remains private until publisher approval. Unpublish and revision restore update public caches correctly. Editors cannot grant themselves publish or administration permissions.

AC20 Search. Given drafts, private records and published articles, when search runs, then only published allowed content is indexed and returned. Query length is bounded; no query is sent to analytics as raw free text. Keyboard selection and pagination work.

AC21 SEO. Given a published page, when inspected without client hydration, then primary content, title, description, canonical and appropriate structured data are present. Sitemap contains only approved indexable 200 routes. Unknown pages return 404.

AC22 Indexing safeguards. Given staging, it requires access control and is noindex. Given production release, automated checks fail deployment for accidental global noindex, broken canonical hostname, malformed sitemap or placeholder company schema.

AC23 Monitoring and errors. Given a simulated intake failure or delayed queue, when alert thresholds are met, then the assigned owner receives an actionable alert with safe request ID. The user sees a recovery message without stack trace or false success. Security contact delivery is verified with an authorized internal test.

AC24 Recovery. Given a restore rehearsal, when database and media are recovered in isolation, then the team measures RPO/RTO, checks representative records and permissions, reapplies deletions and reconciles pending side effects before reopening service.

AC25 Newsletter. Given an unconfirmed email, when signup is requested, then no marketing is sent until the valid confirmation token is used. Expired/reused tokens are handled safely; unsubscribe prevents further campaigns and cannot be overridden by a later unrelated contact enquiry.

AC26 Performance. Given the agreed mobile lab profile and representative pages, when performance tests run, then budgets pass or exceptions are documented before approval. After sufficient field data exists, review mobile and desktop 75th-percentile CWV separately.

57 Edge Cases

Situation	Required behavior
No client projects or testimonials	Omit related sections and links; use process/detail content without fabricated replacements
Unsupported service requested	Accept "Other/Not sure"; route for review without claiming capability
Founder has no company website	Explicit "No website yet" option passes validation
Slow network or lost response	Retain inputs and idempotency key; allow safe retry
Scan takes too long	Explain pending status; allow removal and submission without attachment
CMS unavailable	Serve last approved cached content; intake remains independent
Job closes while form open	Recheck server-side; return clear closure message, not false receipt
Consent revoked mid-session	Stop subsequent optional transmissions; preserve necessary form operation
Long names, Unicode or RTL text entered	Validate by length/purpose, not ASCII-only assumptions; escape output
External URL malicious or internal IP	Store only allowed URL string; never fetch it automatically
Email hard bounce	Mark delivery issue; keep accepted lead visible for authorized follow-up
Duplicate webhook	Unique provider event ID makes processing idempotent
Client removes testimonial permission	Unpublish record and references, invalidate cache and search
Staff leaves company	Revoke identity/session, reassign records, rotate shared secrets if any
Page renamed	Reviewed permanent redirect and updated internal links; no chain
Owner lacks security evidence	Narrow/remove public claim; implement and verify website control separately
Privacy erasure before backup restore	Replay deletion ledger before serving restored data
Analytics blocked	No adverse effect on form; attribution marked unknown rather than guessed

58 Error States

404 copy: “We couldn’t find that page.” Supporting text: “The link may have changed. Explore our services or contact us if you need help.” Links: Home, Services, approved Case Studies, Contact and Search if implemented. Return actual 404 status.

500 copy: “Something went wrong.” Supporting text: “Please try again. If the problem continues, contact [BUSINESS EMAIL] and include reference [SAFE REQUEST ID].” Action: “Try Again” and “Return Home.” Do not imply a form was received when persistence is uncertain; retry with idempotency key.

Form error: “We couldn’t submit your enquiry yet. Your answers are still here. Please try again.” Rate limit: “There have been several attempts. Please wait [RETRY INTERVAL] or contact us by email.”

Attachment rejection: “This file could not be accepted. Use a supported file type within the size limit, or continue without an attachment.” Search empty: “No results found. Try another term or clear filters.”

Offline: “You appear to be offline. Reconnect to submit your enquiry.” Messages must remain visible long enough to act; do not rely on disappearing toasts.

59 Development Roadmap

Planning envelope: approximately 10-14 weeks for an experienced small team, assuming timely business inputs and a selected managed CMS. This is a scheduling assumption, not a supplier quote. Suggested roles: product/content owner, designer, frontend engineer, backend/full-stack engineer, QA and part-time security/DevOps. Some phases overlap; content verification runs throughout.

Phase	Goals and tasks	Deliverables	Dependencies and exit gate
1 Discovery and Requirements	Confirm entity, services, audience, systems and constraints	Approved brief, fact register, risks, PO list	Business owner access; no unidentified launch owner
2 Information Architecture	Map journeys, classify content, verify route scope	79-slot plan adapted to real offer, navigation and URL register	Phase 1; no duplicate-intent or empty destinations
3 Brand and Design System	Select typography, palette, tokens and content tone	Tokens, component principles and sample UI	Phase 1; contrast and licensing checks
4 Wireframes	Design key task flows and page hierarchy	Homepage and key-page wireframes, mobile variants	Phase 2; task walkthrough passes
5 High Fidelity UI	Design real content and all critical states	Responsive designs, prototypes, component variants	Phases 3-4; accessible and owner-reviewed
6 Frontend Foundation	Implement shell, tokens, routing and components	Reusable coded library and representative pages	Phase 5; type/build and responsive checks
7 Backend and CMS	Configure models, permissions, preview, DB and jobs	CMS, migrations, auth and outbox infrastructure	Phases 1-2; role and draft-isolation tests

Phase	Goals and tasks	Deliverables	Dependencies and exit gate
8 Core Pages	Populate approved commercial and corporate content	Publish-ready core pages and metadata	Phases 6-7 plus facts; no placeholders
9 Work and Content System	Implement cases, posts, jobs and taxonomy	Editorial templates, approved initial content	Phase 7; evidence gate and empty states
10 Forms and CRM	Implement intake, uploads, email and routing	Tested contact flow, secure queue, optional CRM	Phase 7; idempotency and outage tests
11 SEO	Validate intent, metadata, schema, redirects and indexing	Crawl report, sitemap, robots and Search Console plan	Phases 8-9; technical SEO gate
12 Security	Threat review, hardening, privacy and recovery	Security review, data map, restore evidence	Phases 7-10; high-risk findings resolved
13 QA	Cross-browser, accessibility, performance and UAT	Test matrix and signed release report	Phases 8-12; critical acceptance criteria pass
14 Production Launch	DNS, release, smoke tests and observation	Live approved release, monitored intake	All P0 gates; tested rollback ready
15 Post Launch Optimization	Review real leads, field performance and content gaps	Prioritized P1 backlog and 30-day review	Phase 14; use measured data, not assumed conversions

Suggested sequence: weeks 1-2 discovery/IA; weeks 2-4 design/content; weeks 4-7 foundation/CMS; weeks 6-9 pages/intake; weeks 9-11 SEO/security/QA; weeks 11-14 release buffer and optimization. Real duration depends on verified content volume, integrations and review speed. P1 quote/search/sector expansion may follow after launch; P2 portal and multilingual work require separate discovery.

60 Launch Checklist

Each item requires an owner, date and evidence link in the release tracker. Unchecked P0 items block launch unless formally re-scoped without leaving misleading UI.

- [] Business owner approves legal name, brand, services, actual delivery model, locations and all public claims.
- [] Every live route has meaningful approved content, a purpose, metadata, contextual links and a next step.
- [] Placeholder/lorem-ipsum/dummy-counter audit is clean across HTML, CMS, public APIs, schema and social previews.
- [] Real case studies, logos, testimonials, staff images and biographies have permission and evidence; absent sections are removed.
- [] Navigation, breadcrumbs, anchor offsets, footer, social URLs and contextual CTAs resolve correctly.
- [] Contact fields, optional values, validation, bot fallback, duplicate retry and success receipt pass end-to-end tests.

- [] Attachments are private, scanned, bounded and authorized; unsupported upload functionality is hidden.
- [] Email sender identity, SPF/DKIM, DMARC plan, reply-to, HTML/plain text, bounce handling and queue alerts work.
- [] CRM failures/retries and fallback lead inbox have been exercised; lead owners know the review process.
- [] Current jobs are real; closed/expired roles reject applications; candidate data permissions pass.
- [] Consent rejection and withdrawal verified using browser network/storage inspection; no PII in analytics.
- [] Event names and conversion definitions validated with test traffic excluded from business reporting.
- [] Counsel approves applicable policies, contractual summaries, retention defaults and privacy wording.
- [] Security contact, support routes and operating-hours statements match actual capability.
- [] Responsive QA covers 320, 375, 390, 430, 768, 1024, 1440 and 1920 widths, long content and zoom.
- [] Keyboard, focus, screen-reader, contrast, labels, error announcements and reduced-motion checks pass.
- [] Browser/device matrix completed and exact versions recorded.
- [] Content rendering and metadata work without client hydration; images have dimensions and meaningful alternatives.
- [] Canonicals, redirects, real 404/500 handling, structured data, sitemap and robots are validated.
- [] Production indexing is enabled intentionally; staging/private environments remain protected.
- [] Performance budgets measured on representative homepage, service, article and form pages.
- [] HTTPS, headers, CSP, staff MFA, permission checks, rate limits, dependency/secret scans and webhook verification pass.
- [] Production secrets are isolated; no customer PII exists in previews or build artifacts.
- [] Backups and point-in-time recovery configured; restore rehearsal measured and documented.
- [] Monitoring detects synthetic intake failure and delayed jobs; primary/backup responders acknowledge test alerts.
- [] DNS, canonical host, certificates, sender DNS, favicon, app icons and approved social preview images work.
- [] Rollback procedure rehearsed and migration compatibility verified; release SHA recorded.
- [] Domain, infrastructure, CMS and integration accounts are owned by the business and access is documented.
- [] Business UAT and QA signoff recorded; known low-risk issues have owners and deadlines.

61 Post Launch Checklist

First day: test a tagged enquiry and permitted job flow; verify durable records, email, CRM and file access; inspect errors and queue age; validate real production metadata and canonical URLs; confirm no accidental optional tracking before consent; monitor certificates/DNS and 404 spikes.

First week: verify Search Console ownership and sitemap processing; review indexing exclusions rather than assuming all pages should index; inspect lead routing and spam; validate analytics against accepted submission counts with consent caveats; fix broken links; verify backups actually ran and alerts reached owners.

First month: review available field performance and lab regressions; analyze qualified leads by approved source aggregates; interview sales about missing buyer information; check privacy deletion process; review content and claim register; assess mobile task completion; confirm current jobs; publish the next substantive resources according to capacity.

Recurring: weekly lead queue and integration review; monthly broken-link/company-contact/backup checks; quarterly security, service capability and permissions review; six-month article refresh review. Changes are prioritized from observed problems, not vanity traffic or speculative A/B tests on insufficient volume. Any experiment requires a stated hypothesis, primary metric, guardrails and adequate sample plan.

62 Documentation Requirements

Developer handover includes a README with project purpose, repository structure and commands; local setup and synthetic fixtures; architecture and trust boundaries; environment variable names/purpose/owner with no secret values; dependency/version policy; API/OpenAPI specification; database migrations and retention map; CMS content model and publishing guide; staff role guide; integration and webhook runbooks; deployment and rollback; backup and recovery; security configuration and incident procedure; monitoring/alert ownership; accessibility test notes; analytics dictionary and consent behavior; SEO/redirect guide; known limitations and maintenance backlog.

Documentation must let a newly authorized engineer run the project, publish a reviewed content change, rotate an integration credential, diagnose a failed enquiry and restore service without relying on the original developer's memory. Keep runbooks in the business-owned repository and provide nontechnical CMS training. Record acceptance of account/repository ownership and credential transfer through a secure channel.

63 Website Due Diligence Readiness Checklist

This is an accuracy and transparency check, not a bank-approval formula. Financial institutions and payment providers apply their own eligibility and risk decisions. Do not change the described business activity to conceal actual services.

- [] Legal identity matches the actual contracting entity and approved registration evidence.
- [] Trading name and legal name relationship is stated clearly where needed.
- [] Registered jurisdiction and public registration details are verified and appropriate to disclose.
- [] Registered/operating addresses are correctly labelled; no virtual address is portrayed as a staffed office.
- [] Website, email domain, contact details, policy identity and genuine social profiles are consistent.
- [] Services describe what the company actually sells and can deliver; no hidden contradictory business activity.
- [] Engagement model explains who contracts, who pays, billing basis and identifiable third-party costs.

- [] No fictitious history, clients, reviews, awards, employees, locations, GitHub activity or certifications.
- [] Case metrics and customer logos are supported by evidence and permissions.
- [] Contact, support and escalation paths work and are monitored.
- [] Privacy, website terms and applicable cancellation/data-processing information are available and reviewed.
- [] Public claims about access, security, support hours and recovery reflect verified actual practices.
- [] Website does not imply regulated services, licences or sector certifications it does not possess.
- [] Ownership and handover explanations align with the actual project agreement.
- [] Private owner identity documents and unnecessary personal details are not publicly exposed.
- [] Navigation and forms function; there are no empty template sections or broken company links.

64 Content Required From Business Owner

Launch blocking company facts

- [] Official legal company name, brand/trading name and relationship between them.
- [] Registration jurisdiction and appropriate public registration number with verification source.
- [] Registered address and separately labelled genuine operating address, if any.
- [] Domain, approved logo files, brand usage and real active social links.
- [] Business email, sales/support contacts, phone and actual monitoring responsibility.
- [] Exact services offered, delivery model, use of employees/contractors and supported locations/timezones.
- [] Verified technology capabilities and an accountable internal reviewer for each.
- [] Contracting entity, billing models, currencies, payment schedule rules and third-party cost treatment.
- [] Actual process, communication cadence, change control, acceptance and handover practices.
- [] IP ownership/licence position, confidentiality process, cancellation/refund terms and escalation route.
- [] Support scope, coverage hours, defect period if any, maintenance options and incident responsibility.
- [] Privacy owner, actual vendors, processing locations, retention and deletion responsibilities.
- [] Applicable policy/counsel review and security practices supported by evidence.

Evidence dependent content

- [] Real founder and team names, roles, approved biographies, photographs and publication permission.
- [] Verified company history and dates; omit timeline if there is nothing meaningful to publish.
- [] Real projects with scope, role, duration, architecture, screenshots and client permission.
- [] Verified outcomes with measurement method, baseline, period and evidence owner.
- [] Genuine testimonials with exact wording and attribution permission.

- [] Partner logos only with actual relationship evidence and permitted use.
- [] Awards, certifications and security attestations only with current evidence and scope.
- [] Actual open jobs, hiring owners, requirements, locations, benefits, compensation disclosures where applicable and recruitment process.
- [] Three or more reviewed initial articles if the homepage resource module will appear.
- [] Rights-cleared images, diagram source files and real screenshots with sensitive information removed.

Operating access and decisions

- [] Business-owned repository, hosting, database, storage, CMS, email and monitoring accounts.
- [] Named product, content, sales, hiring, security and operations owners with backups.
- [] CRM choice, intake assignment rules, email templates and approved response expectations.
- [] Analytics choice, target markets, consent approach and meaningful qualification criteria.
- [] Data region, recovery objectives, retention schedule and approved vendor contracts.
- [] Final P0 service/industry scope and accepted P1/P2 roadmap.

Track inputs in a register: item, owner, supplied date, evidence location, approved wording, approval date and review date. Missing information blocks only the affected claim or feature unless it is essential company identity, contact, privacy or operation. Bracketed placeholders remain in design drafts but never go live.

65 Final Developer Handoff Checklist

- [] All 65 PRD sections are mapped to design files, implementation tickets or explicit scope decisions.
- [] The 79-route/template inventory is reconciled with the published scope; withheld pages have no dangling links.
- [] P0 requirements have assigned owners and trace to AC tests; P1 and P2 are clearly outside launch unless promoted.
- [] Page copy, metadata, FAQs, related links and responsive wireframes are available to the design/content team.
- [] Design tokens and component variants include error, empty, loading, success, keyboard and mobile states.
- [] CMS model, publication gates, evidence register and single-source corporate identity are implemented.
- [] API contracts, schema constraints, role permissions, outbox/idempotency behavior and upload boundaries are agreed.
- [] Lead ownership, email/CRM failure recovery, consent behavior and retention/deletion are operationally tested.
- [] Security, privacy, accessibility, performance and SEO acceptance evidence accompanies the release.
- [] Deployment, backup, restore, rollback and monitoring runbooks are complete and rehearsed.

- [] The business owns accounts, repository, content and applicable deliverables according to the actual contract.
- [] Business owner supplies and approves real facts; counsel approves applicable legal content.
- [] No unsupported company facts from other projects or conversations are inserted into this website.
- [] No claim of bank approval, search ranking, guaranteed revenue, certification or development outcome is made without appropriate basis.
- [] Handoff meeting demonstrates content editing, a real controlled test enquiry, a failed-job retry, permission denial and recovery documentation.

Final specification audit: the planned architecture covers all requested service groups, corporate and trust information, lead handling, content systems and careers. Every page family has content and navigation rules, and all critical dynamic functions have failure behavior and acceptance criteria. Remaining business-specific approvals are deliberately visible in section 64. The live website must prioritize truthful, complete content over the number of pages or appearance of company size.